A SCALAR CONTINUUM THEORY FROM LATTICE YANG–MILLS CORRELATIONS

Abstract. We assemble a scalar continuum hierarchy from uniform separated moments of the Wilson action density. The lattice spacing tends to zero along one admitted bare sequence after infinite-volume Gibbs states have been chosen at each spacing. A quantitative extension theorem controls the resulting moments on Schwartz functions flat on all coincidence diagonals, with derivative order linear in the number of fields and factorial coefficient growth. Reflection positivity passes on the corresponding positive-time test domain. Uniform lattice clustering estimates give realizing pure states for a compact family of continuum phases. The source insertion and physical two-point estimates verify the stated general rotation and reconstruction criteria for these same phases. Their finite means are retained, and the connected short-distance lower bound yields a nonzero scalar excitation. The actual uniform source, running and physical comparison estimates remain hypotheses of the construction.

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1. Introduction

The scalar Wilson action density can be tested at distinct physical points without first defining products at coincident points. Its continuum construction therefore begins with separated smeared correlations. A common subsequence at those correlations is only the first step: reflection positivity, phase selection and reconstruction each use more than pointwise existence of the individual limits.

The required quantitative information is a moment bound uniform in the cutoff and the Gibbs states, with explicit exponential and factorial growth in the number of insertions. We use it first for simultaneous compactness. A product-box Fourier construction then extends the moments to Schwartz tests flat on the coincidence diagonals. Dyadic separation cutoffs and Taylor flatness control the distribution order and its dependence on the number of variables. The positive-time version of the same approximation passes the full reflected form, including its degree-zero terms.

The phase construction uses the all-state estimates before taking continuum limits. The moment estimates control the reflected sizes of local products. The lattice plateau law then gives a physical clustering bound uniformly in the cutoff. Evaluation coordinates provide both the continuum phase law and actual realizing Gibbs-state sequences. This realization allows the source insertion

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and connected two-point estimates to be applied to each phase with the original bare parameters and source normalization.

Ultraviolet stability under explicit running-coupling assumptions is established in Bałaban's [6, Theorem 2 and Corollary 3, pp. 263–264] and [7, Theorem 1, pp. 355–356]. The scalar construction additionally requires the uniform source and physical comparison estimates specified here.

The fixed-spacing inputs are supplied by [1], the trajectory by [2], the graded source and insertion estimates by [3], and the physical short-distance law by [4]. The general plateau and angular continuation results are stated at their actual interfaces below. We check the Euclidean test space, growth and ordered positivity required by the corrected Osterwalder–Schrader theorem [9]. The conclusion concerns the scalar sector generated by the action density. Identifying a complete Yang–Mills observable theory requires additional joint observable constructions.

2. Lattice states and the limiting test spaces

2.1. **The fixed-spacing state space.** Fix an odd integer $L \geq 3$. At spacing $a > 0$, choose one orientation of each bond of $a\mathbb{Z}^4$, and write $\Omega_a = SU(2)^{\mathcal{B}_a}$ with its product topology. The reverse bond carries the inverse matrix. Haar measure has total mass one. For each elementary square p, let U_p be its ordered boundary transport. Its starting vertex and orientation do not affect

$$A_p = 1 - \tfrac{1}{2} \operatorname{tr} U_p, \qquad 0 \leq A_p \leq 2.$$

The trace is real for $SU(2)$. We count each of the six coordinate planes once at each lattice site and write x_p for the plaquette barycentre.

For a finite bond set D, the Wilson specification at $\beta \geq 0$ holds all other bonds fixed and integrates the bonds of D against normalized Haar measure with density proportional to

$$\exp\left[-\beta \sum_{p : p \cap D \neq \varnothing} A_p\right].$$

Here $p \cap D \neq \varnothing$ means that the boundary of p uses a bond of D. A Wilson Gibbs state is a probability measure satisfying these conditional expectation identities. We denote the translation-invariant such states by $\mathcal{G}_{a,\beta}$. A periodic state is one invariant under a finite-index lattice subgroup. The all-mirror theorem of [1, Theorem B] implies that every periodic Wilson state belongs to $\mathcal{G}_{a,\beta}$, is invariant under signed coordinate permutations, and is reflection positive at every coordinate site, coordinate link and diagonal lattice mirror. Its positivity applies to the full bounded Borel half-space bond algebra.

Proposition 2.1 (Fixed-spacing infinite-volume states)**.** For each fixed $a > 0$ and $\beta \geq 0$, periodic extensions of cubic-torus Wilson measures have weak subsequential limits as the torus side tends to infinity. Every such limit lies in $\mathcal{G}_{a,\beta}$. The set $\mathcal{G}_{a,\beta}$ is nonempty, compact and convex in the weak topology. ■

Proof. The countable product Ω_a is compact and metrizable. Its probability measures are therefore weakly sequentially compact. The torus measures, extended periodically, are invariant under every lattice translation. This invariance passes to a weak limit because translations act continuously.

Fix D and a continuous cylinder function F. The conditional kernel $\gamma_D F$ depends on only finitely many bonds and is continuous. To see this, its denominator is at least $e^{-2\beta N(D)}$, where $N(D)$ counts plaquettes meeting D; both numerator and denominator are continuous parameter integrals over a compact group. For sufficiently large tori, the bonds needed by D, its adjacent plaquettes and F embed without identifications. The finite-torus conditional identity is then $\mu_\ell(\gamma_D F) = \mu_\ell(F)$. Weak convergence passes this equality to the limit. Continuous cylinder functions determine measures on the countable compact product, so the identity extends to bounded Borel functions by the monotone class theorem.

The equations for continuous cylinder functions are closed conditions on probability measures. The translation equations are closed too. Hence $\mathcal{G}_{a,\beta}$ is closed in a compact space. Each defining identity is linear in the measure, which proves convexity. □

The proposition keeps a and β fixed. The following cutoff assumptions supply the uniform estimates needed when they vary.

2.2. **One bare sequence and its reference subtraction.** Take $K_j \in \mathbb{Z}_{\geq 0}$, $a_j = L^{-K_j} \downarrow 0$, and nonnegative real bare couplings β_j. The data are assumed to lie on one admitted trajectory family of [2, Theorem A], with the same fixed geometric and source constants. Assume in addition that the ordinary source and actual rooted-domain hypotheses of [3, Theorems B and C] hold uniformly on this family. The short-distance statements will also use the actual running, source-response and comparison assumptions of [4, Theorem A and its cofinal alternative]. These assumptions concern the given lattice theory and the given bare sequence. They do not follow from compactness of $\mathcal{G}_{a,\beta}$.

Choose reference states $\nu_j \in \mathcal{G}_{a_j,\beta_j}$. At each datum, the integrating state $\rho_j \in \mathcal{G}_{a_j,\beta_j}$ may be different. For $f \in C_c^1(\mathbb{R}^4;\mathbb{C})$, define

$$\mathcal{O}_j(f) = \sum_p f(x_p) A_p, \qquad X_j^\nu(f) = \mathcal{O}_j(f) - \nu_j(\mathcal{O}_j(f)). \tag{2.i}$$

The sum is finite. There is no extra power of a_j. This is the Wilson normalization used in [3, 4].

A tuple $(f_1, \dots, f_n)$ is separated if its nonzero entries have pairwise disjoint compact supports. Such supports have positive mutual distance. Set

$$G_{j,n}(f_1, \dots, f_n) = \rho_j\left(\prod_{i=1}^n X_j^\nu(f_i)\right), \qquad G_{j,0} = 1. \tag{2.ii}$$

We use complex multilinearity, not sesquilinearity, for moments. Conjugation enters reflection-positive forms separately. A tuple with a zero entry has value zero. Write

$$\|f\|_\sharp = \max\{\|f\|_\infty, c_M\|\nabla f\|_\infty\}, \qquad \|f\|_1 = \max\{\|f\|_\infty, \|\nabla f\|_\infty\}, \tag{2.iii}$$

where $c_M > 0$ is fixed.

The all-state graded estimate supplied by [3, Theorem B] is the following. There are a fixed integer $k \geq 1$ and $B(R) = C(R+1)^4$ such that

$$|G_{j,n}(f_1, \dots, f_n)| \leq B(R)^n (n!)^k \bar{d}^{-5n} \prod_{i=1}^n \|f_i\|_\sharp. \tag{2.iv}$$

Each support is contained in its own radius-R ball; the ball centres are arbitrary. For $n \geq 2$ the supports have distance at least $d > 0$, and $\bar{d} = \min(d, 1)$. For $n = 1$, set $\bar{d} = 1$. The constants are independent of the cutoff, the two states, the support positions and n. They include complex tests after a fixed enlargement of C. The same estimate applies when $\rho_j = \nu_j$, and when both are replaced by any other pair of states at the same datum. This uniformity will permit phase selection and self-centering without introducing a new continuum sequence.

For each use of a varying constant C, its allowed dependencies are L, c_M, the fixed source and trajectory bounds, and explicitly displayed geometric parameters. Such a constant does not depend on a cutoff, state or field number. Dependence on the field number is displayed through exponential and factorial factors.

2.3. **Flat Schwartz functions.**

Definition 2.2 (The Euclidean test domains)**.** For $n \geq 2$, let $\Delta_n = \bigcup_{1\leq i<h\leq n}\{x_i = x_h\}$ in $(\mathbb{R}^4)^n$, and put

$$\mathcal{S}_{\neq,n} = \{\varphi \in \mathcal{S}(\mathbb{R}^{4n}) : \partial^\alpha \varphi|_{\Delta_n} = 0 \text{ for every multi-index } \alpha\}.$$

Set $\mathcal{S}_{\neq,1} = \mathcal{S}(\mathbb{R}^4)$ and $\mathcal{S}_{\neq,0} = \mathbb{C}$. These spaces carry their Schwartz subspace topology. For r a nonnegative integer and $n \geq 1$, use the seminorm

$$|\varphi|_r = \max_{|\alpha|\leq r} \sup_{x\in\mathbb{R}^{4n}} (1 + |x|^2)^{r/2}|\partial^\alpha \varphi(x)|. \tag{2.v}$$

For $n = 0$, the seminorm is the absolute value. Let $\mathcal{S}_{+,n} \subset \mathcal{S}_{\neq,n}$ consist of functions whose every derivative vanishes whenever at least one time coordinate is nonpositive. Set $\mathcal{S}_{+,0} = \mathbb{C}$. We use the fourth coordinate as time: $x = (\mathbf{x}, t)$, and $\theta(\mathbf{x}, t) = (\mathbf{x}, -t)$. Thus $(\Theta\varphi)(x_1, \dots, x_n) = \varphi(\theta x_1, \dots, \theta x_n)$. ■

These are closed subspaces: every pointwise derivative evaluation is a continuous functional on Schwartz space, and the defining conditions are intersections of their kernels. Tensor products of compactly supported smooth tests with pairwise disjoint supports belong to $\mathcal{S}_{\neq,n}$. Arbitrary tensor products need not belong to it. In particular, continuity on this domain assigns no value to a contact term supported on a coincidence diagonal.

The two components of the construction have different roles. The bound eq. 2.iv gives compactness at every fixed separation and controls the quantitative extension to the flat spaces. The one-field insertion estimate of [3, Theorem C] is an operator estimate uniform in the number of surrounding fields; it is required separately by the angular continuation criterion.

3. A common subsequence for separated tests

Proposition 3.1 (Separated compactness)**.** Let $a_j \to 0$ be the given cutoff sequence. Suppose the moments in eq. 2.ii obey eq. 2.iv for the integrating and centering states (ρ_j, ν_j), and for (ρ_j, ρ_j), with common constants. Assume both states have the lattice translation and signed coordinate-permutation symmetries of the fixed-spacing construction. There is one subsequence on which every moment converges on every tuple of separated compact C^1 tests, at all orders. Write G_n for these reference-centered limits and $G_0 = 1$. They obey the original bound, are multilinear on products of fixed separated support spaces, and are symmetric, real and invariant under all real translations and signed coordinate permutations. Moreover $G_1(f) = b\int f$ for a real constant b. The self-centered limits G_n^0 and all connected limits of order $n \geq 2$ exist on the same subsequence.

For any fixed product of separated compact support neighborhoods, convergence is uniform on its C^1-compact sets of tuples. It also holds for moving tests converging in C^1 with supports in such fixed neighborhoods. Closed inequalities involving finitely many of these moment and connected evaluations pass to the limit, with the same constants when these are uniform. ■

Proof. Consider bounded finite unions of open rational boxes in $\mathbb{R}^4$. They form a countable class. Separated compact supports have neighborhoods U_i in this class whose closures remain pairwise separated, by a finite cover of each support by boxes of sufficiently small diameter. For each such U, choose smooth cutoffs χ_m supported in U and equal to one on the sets $\{x \in U : \operatorname{dist}(x, U^c) \geq 1/m\}$. For example, convolve the indicator of the set at distance greater than $1/(2m)$ with a smooth nonnegative mollifier of mass one and support radius $1/(8m)$.

Use the rational complex spans of χ_m times the real sines and cosines of integral frequencies on integer-period boxes. The resulting family is countable and dense in $C_c^1(U)$ for the C^1 norm. Indeed, for a given test choose $\chi_m = 1$ on a neighborhood of its support. Mollify the test on a smaller scale and periodize it on a box containing $\overline{U}$ and the mollified support in its interior. Uniform continuity of the first derivatives gives C^1 convergence of the mollification. Integration by parts in the Fourier series of the smooth function gives absolute summability after one differentiation. Approximate the

coefficients of a finite partial sum by rational complex numbers and multiply by the fixed cutoff. Multiplication by it is bounded in C^1 and leaves the target test unchanged.

List all finite tuples from these countable families with pairwise separated neighborhood closures, including the one-point tuples. Their values are bounded by eq. 2.iv at their fixed order, radius and separation. Successive subsequences converge at the successive entries of this countable list. Selecting an increasing index from the r-th nested subsequence at stage r gives one sequence converging on the entire list. An eventual bound on each fixed neighborhood product suffices for this selection.

To extend convergence, let $\mathbf{f}, \mathbf{h}$ lie in the same product of fixed separated support neighborhoods. Telescope by replacing the entries one at a time. Every intermediate tuple remains separated. Thus for the fixed order n

$$|G_{j,n}(\mathbf{f}) - G_{j,n}(\mathbf{h})| \le C \sum_{i=1}^{n} \|f_i - h_i\|_\sharp \prod_{l \ne i} \max\{\|f_l\|_\sharp, \|h_l\|_\sharp\}, \tag{3.i}$$

where C depends on n and the radius and separation of the neighborhoods. Approximate a given tuple in its own neighborhoods by a listed tuple. Inequality eq. 3.i makes this approximation uniform in the cutoff tail. The values of the original tuple are Cauchy along the same subsequence. The same inequality proves uniform convergence on compact sets by finite C^1 nets, and convergence on moving tuples by comparison with their limit. It also proves continuity of G_n on each fixed support product. The graded bound passes on the original supports, so its constants use their radius and gap.

Multilinearity on these products, symmetry and reality pass from the lattice identities. For a real translation v, choose $v_j \in a_j\mathbb{Z}^4$ with $v_j \to v$. For $(\tau_v f)(x) = f(x - v)$ one has

$$\|\tau_{v_j} f - \tau_v f\|_\sharp \le \max\{\|\nabla f\|_\infty |v_j - v|, c_M \omega_{\nabla f}(|v_j - v|)\},$$

where $\omega_{\nabla f}$ is the uniform modulus of continuity of ∇f. Fixed separated neighborhoods of the limiting translated supports contain the moving supports for all large j. Lattice translation invariance and moving-test convergence give invariance under v. The signed coordinate permutations act exactly at each cutoff. The same moving-test conclusion applies to pullbacks by converging Euclidean isometries whenever the limiting tuple remains separated, by the chain rule and uniform continuity.

At one point the difference of the two plaquette means is the same number δ_j for all positions and planes. This uses the signed coordinate permutations as well as translations. Put

$$b_j = 6a_j^{-4}\delta_j, \qquad R_j(f) = \frac{a_j^4}{6} \sum_p f(x_p), \qquad G_{j,1}(f) = b_j R_j(f). \tag{3.ii}$$

For each of the six plane labels the shifted grid gives a Riemann sum for $\int f$. The mesh cubes meeting a support of radius R have total volume at most $C(R + 2a_j)^4$. The oscillation in one cube is at most $2a_j\|\nabla f\|_\infty$, so

$$|R_j(f) - \textstyle\int f| \le C(R + 2a_j)^4 a_j \|\nabla f\|_\infty.$$

Choose a fixed compact test of integral one and radius at most one. Its Riemann sum is at least $1/2$ eventually, independently of the two states. The one-point bound then bounds b_j uniformly over the states. Convergence at this test gives $b_j \to b \in \mathbb{R}$ along the chosen subsequence. The same identity then gives $G_1(f) = b \int f$ for every test.

For a finite index set write $\mathbf{f}_A = (f_i)_{i \in A}$ in its inherited order. Self-centering is obtained by subtracting $G_{j,1}(f_i)$ from each reference-centered variable. Expanding the product and passing to the limit gives

$$G_n^0(\mathbf{f}) = \sum_{A \subseteq \{1,\dots,n\}} G_{|A|}(\mathbf{f}_A) \prod_{i \notin A} \left(-b \int f_i\right). \tag{3.iii}$$

Its graded bound passes directly from the bound for (ρ_j, ρ_j). In particular $G_0^0 = 1$ and $G_1^0 = 0$. For $n \geq 2$ the cumulant is unchanged by deterministic shifts. Its limit is the finite partition sum

$$S_n^T(\mathbf{f}) = \sum_{\pi} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{A \in \pi} G_{|A|}(\mathbf{f}_A). \tag{3.iv}$$

Here π ranges over partitions of $\{1, \dots, n\}$. The formula follows from expanding the logarithm of the normalized source generating function at zero, or equivalently the moment–cumulant identity of [3, Lemma 2.1]. In particular G_n^0 is the sum of products of the S_l^T over partitions with all block sizes at least two.

The same factorial exponent works for the connected functions. Indeed we may take the exponent k of eq. 2.iv to be at least one. The moment weights of a partition into sizes $n_1, \dots, n_r$ carry $\prod_l (n_l!)^k \leq (n!)^{k-1} \prod_l n_l!$. Summing $\prod_l n_l!$ over ordered blocks of any one composition of n gives $n!$, since the number of assignments is $n! / \prod_l n_l!$. Each unordered partition with r blocks occurs $r!$ times. The sum weighted by $(r-1)!$ is therefore $n! \binom{n-1}{r-1} / r$. Summing over r gives at most $2^{n-1} (n!)^k$ after the common $(n!)^{k-1}$ factor. The remaining moment bounds multiply to the original product norm and separation power. Hence one extra factor 2^n in the exponential constant suffices.

All the limits used here are simultaneous on each fixed finite collection of tuples and their subtuples. Thus inequalities defined by closed conditions on such evaluations, with any convergent scalar coefficients, pass by continuity. This also applies to moving tuples by eq. 3.i. □

Corollary 3.2 (Torus sequences)**.** Consider compatible cubic Wilson tori with $a_j \to 0$ and even bare lattice side counts ℓ_j / a_j. Use their Wilson expectation for both integration and centering. On a fixed torus, uniform bounds of the form $C_{n,R,d} \prod_i \|f_i\|_\sharp$ at each order and support radius and separation suffice for one subsequence on all separated C^1 tuples. The limiting hierarchy is self-centered and has all torus translations and signed coordinate-permutation symmetries. If the torus sides tend to infinity, a fixed compact tuple is transported in a common injective lift. Under the finite-torus graded bounds with uniform constants, the same construction gives the separated limits and bounds of Proposition 3.1 on $\mathbb{R}^4$.

Suppose instead that a self-centered continuum hierarchy has already been extracted at each of a sequence of torus sides tending to infinity. A separated limit of those hierarchies, with the uniform per-support bounds at finite cutoffs, is also realized by one sequence of original torus data with meshes tending to zero. ■□

Proof. On a fixed torus use a countable basis of rational coordinate boxes in the period-normalized coordinates, the same support-adapted cutoffs and rational Fourier polynomials. The preceding extraction and telescoping proof applies there. Translations are rounded modulo the period. The one-point functions vanish because the tori use self-centering. On growing tori any fixed finite family of compact support neighborhoods fits, eventually, in a common lift of diameter less than a quarter of the side. Periodizing the tests in this lift preserves the support distances and C^1 norms. Indeed the copies are disjoint, and the quotient is an isometry on the chosen lift. Thus the bounds needed for each entry of the countable list hold eventually. The moving-test and symmetry arguments then apply on the lifted support neighborhoods with their fixed margins.

For the last assertion fix the countable list of separated Euclidean tuples used above. On the j-th torus select one term of its defining cutoff subsequence with $K \geq j$ whose values differ by at most $1/j$ from the fixed-torus limits on the first j listed tuples whose neighborhoods embed there. There are finitely many such conditions, and each holds eventually along the defining subsequence at that torus. Every fixed listed tuple eventually embeds and is among those tested. The selected torus values and the iterated limits therefore agree on this list. Inequality eq. 3.i with the common local constants applies both to the inner limits and the selected finite cutoffs. Approximate any fixed tuple in its separated neighborhoods, and let this approximation error tend to zero after the diagonal limit. This gives agreement on every fixed separated tuple and all its subtuples. □

3.1. **Recognition on sufficiently large tori.**

Proposition 3.3 (Large torus recognition)**.** For each fixed bare datum $K \in \mathbb{Z}_{\geq 0}$, $\beta \geq 0$, with $a_K = L^{-K}$, fix one compatible cofinal family of Wilson tori and let $\mathcal{G}^{\mathrm{tor}}_{K,\beta}$ be the set of weak limit points of their periodic lifts. Write μ_{torus} for the periodic lift at the compared side. There is a side threshold $\ell_\flat(K,\beta) \geq 2L^K$ with the following properties. For every side in that family above the threshold there is a state $\mu \in \mathcal{G}^{\mathrm{tor}}_{K,\beta}$ whose marginal on the bonds of the physical max-norm cube of radius $K+1$ has Haar density ratio to the torus marginal within $\epsilon_K = 2e^{-1/a_K}$ of one uniformly.

For $K \to \infty$ on such sides, replacing each torus law by its chosen state leaves the limiting connected correlations and self-centered moments of every fixed separated compact tuple unchanged. Conversely, a sequence of states in these torus limit sets at their respective bare data admits torus approximants above the same thresholds with the same limits. On the chosen cube the torus diagonal reflection form satisfies

$$\mu_{\mathrm{torus}}(\Theta_\Pi \overline{F}\, F) \geq -\epsilon_K\, \mu_{\mathrm{torus}}(|F|^2) \tag{3.v}$$

for a bounded Borel F on one side of a diagonal mirror Π through the origin. Here Θ_Π is the linear pullback by the bond reflection; conjugation is written separately. ∎

Proof. At fixed spacing first use any finite bond set W such that it and the exterior bonds of its incident plaquettes embed in the torus. Write $h_W(V \mid U)$ for the conditional density in the Wilson specification of Proposition 2.1, relative to normalized Haar on W. It is continuous in the finitely many interior and exterior variables. If N_W is the number of plaquettes meeting W, then $h_W \geq e^{-2\beta N_W}$, since every plaquette action lies in $[0,2]$. For the periodic lift $\widetilde{\mu}_\ell$ and each limit state μ the respective marginals have densities

$$m_{\widetilde{\mu}_\ell}(V) = \int h_W(V \mid U)\, d\widetilde{\mu}_\ell(U), \qquad m_\mu(V) = \int h_W(V \mid U)\, d\mu(U).$$

The first formula follows by integrating the embedded window in the finite Wilson integral. The second is the DLR identity. The family $h_W(V \mid \cdot)$ has compact image in the sup norm as V ranges over a compact product of copies of $SU(2)$. Thus along any subsequence of torus lifts converging weakly to μ, a finite sup-norm net of this family gives uniform convergence of their marginal densities to m_μ. The common positive lower bound gives uniform convergence of $m_\mu / m_{\widetilde{\mu}_\ell}$ to one on that subsequence. It follows that

$$\inf_{\mu \in \mathcal{G}^{\mathrm{tor}}_{K,\beta}} \| m_\mu / m_{\widetilde{\mu}_\ell} - 1 \|_\infty \longrightarrow 0 \qquad (\ell \to \infty).$$

Otherwise a violating sequence of sides would have a weak subsequence with a limit in this same set.

Now take all bonds with both endpoints in the radius-$(K+1)$ cube, with either orientation represented once. Choose $\ell_\flat$ so that the window with its plaquette collar embeds and the last infimum is at most e^{-1/a_K} on every later side. Increasing it to $2L^K$ if needed gives the stipulated growth. Select one state within $2e^{-1/a_K}$ in the density-ratio norm. For every bounded measurable B of the window,

$$|\mu(B) - \widetilde{\mu}_\ell(B)| \leq \epsilon_K\, \widetilde{\mu}_\ell(|B|). \tag{3.vi}$$

This is one window estimate for all the observables used below.

Fix finitely many compact tests in a cube of physical radius R, with a plaquette margin inside the window for large K. The number of plaquette centres per test is at most $C(1+R/a_K)^4$. Hence eq. 3.vi bounds a difference of raw moments of order n by

$$C^n \epsilon_K (1+R/a_K)^{4n} \prod_i \|f_i\|_\infty.$$

Expansion over subsets for self-centering and over partitions for cumulants gives the same estimate with a constant depending on the fixed order. For a product of expectations, replace its factors

one at a time and use the corresponding raw bounds for the other factors. The exponential e^{-1/a_K} dominates every fixed power of a_K^{-1}. This proves recognition of all fixed-tuple limits, with the same chosen state for every tuple eventually contained in the window. For the converse, at each fixed datum use a cofinal subsequence converging to its prescribed torus limit point. The ratio convergence already proved allows a side as large as desired at that datum with the same window tolerance. The same moment estimate gives the converse.

For the reflection claim, the cube and its bond set are invariant under a diagonal reflection through the origin. The torus lift is invariant under this signed permutation, so its expectation of $\Theta_\Pi \overline{F}\, F$ is real. The state μ is reflection positive by [1, Theorem B]. Apply eq. 3.vi to $\Theta_\Pi \overline{F}\, F$ and use

$$\widetilde{\mu}_\ell(|\Theta_\Pi \overline{F}|\,|F|) \le \widetilde{\mu}_\ell(|F|^2)$$

by Cauchy–Schwarz and reflection invariance. This gives eq. 3.v. Its error tends to zero for polynomials of fixed degree in fixed compactly supported observables and their centered versions, by the same plaquette count. For an affine lattice diagonal mirror, a lattice translation moves it to the origin. On fixed bounded support sets and bounded mirror offsets the translated observables lie in the same growing window eventually. □

4. Quantitative extension to flat Schwartz tests

Use the seminorm eq. 2.v and the spaces of Definition 2.2, with the one-slot norm $\|f\|_1$ fixed in eq. 2.iii.

Proposition 4.1 (Flat extension and separated tensor density)**.** Let F_n, $n \ge 1$, be forms on separated compact C^1 tuples, multilinear on each product of separated support spaces. Suppose $D \ge 1$, $\kappa \ge 0$ and $\alpha \ge 0$ are independent of n, and for radius-one supports at mutual distance at least $d > 0$ one has

$$|F_n(f_1, \dots, f_n)| \le D^n (n!)^\kappa \min(1, d)^{-\alpha n} \prod_i \|f_i\|_1. \tag{4.i}$$

At $n = 1$ take $d = 1$. Set $A = \lceil \alpha \rceil$ and $k_* = A + 15$. There is a unique continuous linear extension to $\mathcal{S}_{\neq,n}$ agreeing with the values on smooth compact separated tensors, and

$$|F_n(\varphi)| \le C^n (n!)^{\kappa+32} |\varphi|_{k_* n}, \qquad n \ge 1. \tag{4.ii}$$

Here C depends on D, α and numerical cutoff choices, uniformly in n and all families obeying eq. 4.i with those constants. Finite sums of the indicated tensors are dense in $\mathcal{S}_{\neq,n}$ in the Schwartz topology. For the positive-time flat subspace, their entries may all have compact support strictly in the open positive half-space. This density assertion also holds for any affine half-space by an isometric change of test coordinates. The radius-one bound also gives a radius-R bound in eq. 4.i with $CD(1 + R)^4$ in place of D. ■

Proof. Cutoffs. We use a finite set of fixed smooth cutoff functions valued in $[0, 1]$ whose derivatives of order q are bounded by $C^{q+1}(q!)^2$. To construct them, start with $a(t) = e^{-1/t}$ for $t > 0$ and zero for $t \le 0$. On the complex circle $|z - t| = t/2$ one has $\mathrm{Re}(1/z) \ge 2/(9t)$. Cauchy's formula gives

$$|a^{(q)}(t)| \le q!(2/t)^q e^{-2/(9t)}.$$

The maximum of $t^{-q} e^{-c/t}$ and the inequality $(q/e)^q \le q!$ give the stated bounds and flatness at zero. The smooth step $s(t) = a(t)/(a(t) + a(1 - t))$ is zero at $t \le 0$ and one at $t \ge 1$. On $[0, 1]$ its denominator is at least e^{-2}. A reciprocal of a denominator bounded below by $\delta > 0$ has the same derivative bound. Indeed, differentiating $DD^{-1} = 1$ and dividing by $(q!)^2$ introduces the coefficients $1/\binom{q}{r}$. If the denominator bound uses a constant C_D, the induction for the reciprocal closes with $MB^q(q!)^2$, $M \ge \delta^{-1}$, as soon as $(C_D/\delta) \sum_{r \ge 1} (C_D/B)^r \le 1$. Leibniz's rule then gives the bound for the step.

Take an even cutoff $\eta(t)$ equal to one for $|t| \le 5/4$ and zero for $|t| \ge 2$, using $s((2-|t|)/(3/4))$ on the transition. Put $\psi(t) = \eta(2t)$ and $\chi_1(t) = \psi(t)/\sum_{m\in\mathbb{Z}} \psi(t-m)$. The denominator is at least one by the nearest integer, and its locally finite sum uses at most three terms and has the same derivative bounds. Define on $\mathbb{R}^4$

$$\chi(u) = \prod_{\mu=1}^{4} \chi_1(u_\mu), \qquad \chi^+(u) = \prod_{\mu=1}^{4} \eta(u_\mu), \qquad \zeta(u) = 1 - \chi^+(u).$$

The integer translates of χ sum to one. Its support is in $[-1,1]^4$, where $\chi^+ = 1$, and χ^+ has support in $[-2,2]^4$. Thus ζ is zero on the first cube and one outside the second. All these cutoffs have the stated derivative bounds. For N such factors at a scale proportional to h, Leibniz's rule yields

$$\left| \partial^\beta \prod_{\nu=1}^{N} b_\nu \right| \le (CN)^q (q!)^2 h^{-q}, \qquad q = |\beta|. \tag{4.iii}$$

To see the dependence on N, distribute the derivative among the factors. The multinomial coefficients sum to N^q. The assigned factorials have product at most $q!$, and at most q factors are differentiated. Each undifferentiated factor is bounded by one. Using a difference $x_i - x_l$ as an argument changes only derivative signs in this estimate.

For the radius-R assertion, use translates of this partition at a fixed mesh small enough for each piece to fit in a unit ball. At most $C(1+R)^4$ pieces meet a support of radius R. Their C^1 norms are bounded by a fixed multiple of the original norm. Multiplication by a cutoff preserves containment in the original support, and thus preserves each interslot gap. Multilinearity and the radius-one estimate give the assertion.

Local product boxes. Let $0 < h \le 1/16$ and $\lambda = h/64$. Suppose the cubes $c_i + [-2\lambda, 2\lambda]^4$ have mutual distance at least $h/2$. For u smooth and compactly supported in the product of open inner cubes $c_i + (-5\lambda/4, 5\lambda/4)^4$, expand it on the product of the period-8λ cubes. Multiply each one-slot Fourier exponential by $\chi^+((x_i - c_i)/\lambda)$. The resulting product series equals u, since the periodic extension agrees with u on the support of the product cutoff. For its Fourier coefficients $\widehat{u}(k)$, $k = (k_i)_{i=1}^n$, put

$$J_{r,\lambda}(u) = \max_{|\beta|\le r} \lambda^{|\beta|} \|\partial^\beta u\|_\infty.$$

Rescale each slot to a period-two cube. Integration by parts with $(1 - \pi^{-2}\Delta_i)^4$ in each slot bounds the coefficients by

$$|\widehat{u}(k)| \le C^n J_{8n,\lambda}(u) \prod_i (1 + |k_i|^2)^{-4}.$$

The coefficient sum of this differential operator is a fixed constant per slot. The Fourier normalization is the inverse volume of the period cube, so it does not add a power of λ^{-1}. Each localized exponential has $\|\cdot\|_1 \le C\lambda^{-1}(1+|k_i|)$. The single-slot sum of $(1+|k_i|)(1+|k_i|^2)^{-4}$ over $\mathbb{Z}^4$ is finite. Equation 4.i therefore defines a local functional $T(u)$ by an absolutely convergent series and gives

$$|T(u)| \le (CD)^n (n!)^\kappa h^{-(\alpha+1)n} J_{8n,\lambda}(u). \tag{4.iv}$$

For a tensor in the inner cubes the Fourier coefficients factor. The localized one-slot series converge in C^1 on fixed separated support spaces. The multilinear bound shows that the local functional takes the prescribed tensor value. On overlaps, partition a compactly supported test into smaller product boxes whose expanded supports are inside the overlap. This is possible since the overlap is open and disjoint from the diagonals. In each smaller box the same cutoff Fourier series converges with all derivatives on a fixed compact set, by integration by parts at any fixed order. Both local functionals agree on these tensor approximants and are continuous by eq. 4.iv, so they agree on the test. Finite smooth partitions of unity now define a single functional on compact tests off the diagonals. A different partition gives the same value by refining both partitions and using agreement on overlaps. For an arbitrary original separated smooth tensor, partition each slot

into smaller compact pieces with their expanded boxes still separated across slots. Multilinearity on the original support product then gives the prescribed value. In particular linear relations among separated tensors have zero value under this construction.

Summing over scales and positions. Let $h_j = 2^{-j}/16$, $\lambda_j = h_j/64$, and set

$$P_h(x) = \prod_{i<l} \zeta((x_i - x_l)/h), \qquad D_0 = P_{h_0}, \qquad D_j = P_{h_j} - P_{h_{j-1}} \quad (j \geq 1).$$

Off the diagonals the sum of the D_j is one. A fine layer and all its derivatives are supported where the smallest max-norm pair distance lies between h_j and $4h_j$. At $n = 1$ the pair product is one, so there are no fine layers. At level j use the product partition $\chi_{j,z}(x) = \prod_i \chi(x_i/\lambda_j - z_i)$ for $z = (z_i)_{i=1}^n \in (\mathbb{Z}^4)^n$. When $\chi_{j,z} D_j$ is nonzero, a point in its support has every pair distance at least h_j. The slot centres are at pair distance at least $h_j - 2\lambda_j$ in max norm. Their expanded cubes are separated by at least $h_j - 6\lambda_j > h_j/2$. The closed grid supports lie strictly inside the open inner cubes used in the local construction. Thus eq. 4.iv applies to $u = \chi_{j,z} D_j \varphi$.

In each layer there are at most Cn^2 cutoff factors. For derivatives through order $8n$, eq. 4.iii and the scale factor in J_{8n,λ_j} give a cutoff cost bounded by

$$(Cn^2)^{8n}((8n)!)^2 \leq C^n (n!)^{32}. \tag{4.v}$$

Here $n^{16n} \leq e^{16n}(n!)^{16}$, and $(8n)! \leq 8^{8n}(n!)^8$ by the multinomial formula. No dimension-dependent constant is used in place of this bound.

On a fine layer move the first point of a closest pair onto the second. The resulting point is on a diagonal at Euclidean distance at most $8h_j$. The segment uses at most four coordinate directions. Since all derivatives of φ vanish at its diagonal endpoint, Taylor's formula gives, for $|\beta| \leq 8n$ and $m = (A+5)n + 2$,

$$|\partial^\beta \varphi(x)| \leq C_A^n h_j^m (1 + |x|^2)^{-k_* n/2} |\varphi|_{k_* n}.$$

Indeed $8n + m \leq k_* n$. The directional remainder has coefficient at most $(Ch_j)^m/m!$, using the four-coordinate ℓ^1 displacement. The weights along a segment of length at most $1/2$ are comparable to the weight at x, at cost $C^{k_* n}$. Both costs are bounded by C_A^n after retaining the displayed power of h_j. For $j = 0$ use the direct seminorm bound and omit h_j^m.

By Leibniz's rule, the scale factor in J_{8n,λ_j} cancels the inverse powers of h_j on the cutoff derivatives. Combining the last estimates gives

$$J_{8n,\lambda_j}(\chi_{j,z} D_j \varphi) \leq C_A^n (n!)^{32} h_j^m |\varphi|_{k_* n} \sup_{x \in \operatorname{supp} \chi_{j,z}} (1 + |x|^2)^{-k_* n/2}$$

on a fine layer, with the same coarse-layer convention. For the grid sum use $1 + |x|^2 \geq \prod_i (1 + |x_i|^2)^{1/n}$. Each x_i is a bounded distance from its grid centre, so

$$\sum_z \sup_{x \in \operatorname{supp} \chi_{j,z}} (1 + |x|^2)^{-k_* n/2} \leq C_A^n \prod_i \sum_{z_i \in \mathbb{Z}^4} (1 + |\lambda_j z_i|^2)^{-k_*/2} \leq C_A^n h_j^{-4n}.$$

The one-slot bound follows by integrating over its mesh cubes, since $k_* > 4$. Substitution in eq. 4.iv gives the fine-scale bound

$$\sum_z |T(\chi_{j,z} D_j \varphi)| \leq C^n (n!)^{\kappa + 32} |\varphi|_{k_* n} h_j^{m - (\alpha+5)n}.$$

The power of h_j is at least two. At $j = 0$ its fixed scale cost is absorbed in C^n. The sum in j is absolute, so it defines the extension and proves eq. 4.ii. For compact support away from the diagonals this sum is finite and is the glued local functional.

Density and the time boundary. First multiply by $\prod_i \chi^+(x_i/R)$ and send $R \to \infty$. For a fixed seminorm, the discarded derivatives have a point with $|x_i| \geq R$, and one extra power of Schwartz decay bounds their seminorm by $C_{n,r} R^{-1} |\varphi|_{r+1}$. Next multiply by P_h and send $h \downarrow 0$. Every surviving derivative of $(1 - P_h)\varphi$ has a pair within max-norm distance $2h$. For fixed n, r, the

preceding Taylor argument with any $M > r$ gives a bound $C_{n,r,M} h^{M-r} |\varphi|_{r+M}$. Thus compact tests away from the diagonals are dense. Apply a finite product-box partition and the cutoff Fourier expansions to those tests. Partial sums converge with all derivatives on a common compact set, hence in the Schwartz topology. Choosing the cutoffs and finite sums successively for the first r seminorms gives a single sequence of finite sums of separated tensors.

On $\mathcal{S}_{+,n}$ also multiply by $B_h(x) = \prod_i s(x_{i,4}/h - 1)$. The cutoff is zero when a time is at most h and one when all are at least $2h$. Where it removes a nonzero derivative of a positive-time test, some time lies in $[0, 2h]$. Taylor expansion to that time hyperplane gives the same decay h^{M-r}, because every jet there vanishes. The compact supports of the approximants lie strictly in the positive half-space, so their enlarged Fourier boxes can be chosen there too. If one uses $P_h B_h$ directly in the quantitative dyadic construction, a fine layer has one close pair or one time within $4h$ of zero. The same Taylor and cutoff estimates apply, since there are still at most Cn^2 factors. The constants in eq. 4.ii retain the same order and factorial exponent on that subspace. An affine isometry of the tests gives the density statement for another half-space.

Continuity and this density give uniqueness, independently of the boxes and cutoffs. They also transport any symmetry of the separated forms whose test pullback is continuous in the Schwartz topology. □

Lemma 4.2 (Integration of omitted slots)**.** For $A \subseteq \{1, \dots, n\}$, let I_A integrate the variables indexed by A, leaving the other slots in order. It maps $\mathcal{S}_{\neq,n}$ continuously into $\mathcal{S}_{\neq,n-|A|}$ and preserves positive-time flatness in the remaining variables. If $m = |A|$ and $r \geq 0$ is an integer, then

$$|I_A \varphi|_r \leq C_6^m |\varphi|_{r+6m}, \qquad C_6 = \int_{\mathbb{R}^4} (1 + |y|^2)^{-3} \, dy. \tag{4.vi}$$

For no remaining variables the left side is the absolute value of the scalar integral. ■

Proof. Let x collect the remaining variables and y_i the integrated ones. The weight inequality

$$(1 + |x|^2)^{r/2} \prod_i (1 + |y_i|^2)^3 \leq (1 + |x|^2 + \textstyle\sum_i |y_i|^2)^{(r+6m)/2}$$

gives an integrable majorant for every retained derivative of order at most r. Differentiation under the integral and integration of the product weights yield eq. 4.vi. On a remaining diagonal or a forbidden retained time, the corresponding derivatives vanish at every integrated datum. They therefore vanish after integration as well. □

Corollary 4.3 (Growth and convergence of the extended hierarchy)**.** The separated limits of Proposition 3.1 extend uniquely to the flat spaces. Their full and self-centered moments, and their higher connected functions, satisfy

$$|F_n(\varphi)| \leq C^n (n!)^{k+32} |\varphi|_{20n}, \qquad n \geq 1, \tag{4.vii}$$

uniformly for limits from the same graded input constants. For infinite-lattice sequences the same subsequence converges on every flat Schwartz test after extending the cutoff forms. For growing tori the conclusion concerns the extended limit of the compact tuples. The fixed-torus limits define unique distributions off the coincidence diagonals. ■□

Proof. On radius-one supports $\|f\|_\sharp \leq \max(1, c_M) \|f\|_1$. The graded input and the cumulant estimate therefore give eq. 4.i with $\alpha = 5, \kappa = k$ and a common D, after increasing it by a fixed factor. Proposition 4.1 gives eq. 4.vii. It applies with the same constants to any admitted phase realized on a sequence of states with the same two-state estimates, including its reference-centered one-point function.

For the infinite-lattice sequence the uniform graded bound also applies to each cutoff form. Approximate a fixed φ by a finite sum of smooth separated tensors in $|\cdot|_{20n}$. On that finite sum the cutoff limits are already known. Equation 4.vii bounds the approximation error at both ends

uniformly, so the convergence extends to φ. On growing tori the support-lift argument applies eventually to each compact tuple. This suffices for the bound on the limit. At a finite torus distant Euclidean supports may overlap after projection, so a global Euclidean cutoff functional is not used there.

For a fixed torus take smooth cutoffs in a product of separated coordinate neighborhoods. Expand a joint compact test in torus Fourier modes and localize each slot by its cutoff. At fixed n the per-support moment bound costs at most $C(1+|k|)^n$ on a mode of combined frequency $k \in \mathbb{Z}^{4n}$. Integrating by parts $5n+1$ times in a coordinate of maximal frequency gives a summable bound after this loss, since the remaining exponent is $4n+1$. The series defines a local functional of finite order. It agrees with the tuple moments by the same separated tensor approximation as in the product-box proof. Agreement on overlaps and a partition of unity give the unique off-diagonal distribution. □

5. Reflection forms and insertions

Use one subsequence of the two-state moments eq. 2.ii. In this section G_n denotes its limit with the given reference subtraction. The test spaces and Schwartz seminorms are those in Definition 2.2. For an affine hyperplane Π let θ_Π be reflection in Π, and let Θ_Π denote simultaneous pullback on every test variable. Choose a positive side of Π. The space $\mathcal{S}_{+,n}(\Pi)$ consists of diagonal-flat Schwartz tests also flat outside the set in which all variables lie in that open side. Its degree-zero term is $\mathbb{C}$. We write $\tau_v f(x) = f(x-v)$, with the simultaneous action on multivariable tests. For $\Pi = \{x_4 = 0\}$ the positive side is $x_4 > 0$.

Proposition 5.1 (Reflection positivity of the limiting moments)**.** Suppose the two-state moments converge on the separated tuples as in Proposition 3.1. Let their diagonal-flat extensions be those of Proposition 4.1. For every affine coordinate or diagonal mirror Π, and every finite sequence $F = (F_0, \dots, F_N)$ with $F_n \in \mathcal{S}_{+,n}(\Pi)$,

$$q_{G,\Pi}(F,F) := \sum_{m,n=0}^{N} G_{m+n}(\Theta_\Pi \overline{F_m} \otimes F_n) \geq 0. \tag{5.i}$$

The same formula with a second sequence in the second slot defines a Hermitian positive semidefinite form, antilinear in the first sequence. It is continuous at each pair of fixed finite degree bounds in the induced Schwartz topologies. The completion of its quotient by the null space is separable. ■

Proof. First take finite linear combinations of compact smooth product tests. In each product the factors have mutually separated supports on the positive side. There is no separation requirement between supports belonging to different products. Choose lattice mirrors Π_j of the same normal type with heights tending to that of Π. Coordinate site and link planes and diagonal planes are supplied by [1, Theorem B]. Every contributing plaquette in a positive product is wholly on the positive side of Π_j for large j, since its diameter is $O(a_j)$ and the support has a fixed positive distance to Π. Thus the corresponding polynomial in X_j^ν is a bounded function of the positive-side bonds at the fixed cutoff.

The barycentre of a reflected plaquette is the reflected barycentre. Its trace is unchanged by a change of starting vertex or orientation. Consequently the raw observable intertwines the lattice reflection and pullback of the test. Invariance of ν_j under Π_j gives the same identity for the subtracted scalar. Apply lattice reflection positivity in ρ_j to the entire polynomial, including its constant term. Each entry of the resulting form is a moment of a separated tuple: the left entries lie on the negative side, the right entries on the positive side, and each side is internally separated. Pullback by θ_{Π_j} converges to pullback by θ_Π in C^1 on a fixed compact neighbourhood of each reflected support. The moving-test conclusion of Proposition 3.1 therefore passes the finite sum to eq. 5.i. Its degree-zero entry is $G_0 = 1$.

For arbitrary F_m, F_n in the stated spaces, their mixed tensor is diagonal-flat. At a coincidence within one group every differentiated term vanishes by flatness of that group. At a coincidence across groups, the common point cannot lie strictly on the respective open sides for both factors. Every derivative of at least one factor is therefore zero at that point. The same argument covers zero time and all multiderivatives. Affine pullback, conjugation and tensor product are continuous in Schwartz topology. For example, the product of the two weights of order r bounds the joint weight of order r. Thus the continuous extensions of G_{m+n} apply to these mixed tensors. Approximate each degree by finite sums of the separated positive-side tensors of Proposition 4.1, after an isometry taking Π to the coordinate plane. Passing to the limit in the finitely many terms gives the claimed inequality. Polarization gives Hermiticity and Cauchy–Schwarz. Schwartz space is separable and metrizable, and each flat subspace is a separable subspace. Each finite direct sum therefore has a countable dense set. Continuity of the form makes its image dense for the quotient norm. The union of these sets over the finite degree bounds is countable and dense in the Hilbert completion. This proves separability. □

Corollary 5.2 (Reflection positivity of growing-torus limits)**.** Use the direct growing-torus compactness of Corollary 3.2, with the graded bounds and diagonal-flat extension of Proposition 4.1. Then eq. 5.i holds on every affine coordinate mirror for the self-centered limit. If the torus sides also lie above the thresholds of Proposition 3.3, it holds on every affine diagonal mirror as well. ■□

Proof. Begin with the compact separated positive-side polynomials in Proposition 5.1. Round a coordinate mirror to lattice mirrors of one fixed type. The antipodal planes have the same type, since the number of lattice sites along each side is even. For large j the compact positive supports and their sampled plaquettes lie strictly within the torus half between these two planes. The first plane converges to the prescribed mirror and the antipodal plane leaves every fixed support. Finite-torus reflection positivity from [1, Proposition 3.1] gives the nonnegative form on the entire lattice polynomial. Invariance makes torus self-centering commute with reflection. Use the fixed common lift of the supports and the moving-test convergence to pass each form entry.

For a diagonal plane through the origin use the approximate lattice inequality eq. 3.v. For a fixed polynomial of degree at most N, the supremum norm of the centered lattice polynomial is bounded by Ca_j^{-4N} for all sufficiently small a_j. This uses its fixed compact supports and the plaquette count. Thus $\epsilon_{K_j}\mu_{\mathrm{torus}}(|F_j|^2) \to 0$. The limiting form is nonnegative. Translation invariance carries it to every affine mirror of that orientation. In both cases the separated tensor density and continuity argument from Proposition 5.1 gives the full diagonal-flat forms. □

For a mirror Π, let $\mathcal{H}_{G,\Pi}$ be the completion of the finite-sequence quotient by the null space of $q_{G,\Pi}$. Write $[F]_G$ for a class when the mirror is understood. The degree-zero class Ω has norm one. Compact separated positive-side tensor polynomials are dense in this quotient. One can also use compact C^1 factors: smooth each within its fixed separated support neighbourhood, and use the graded bound for the differences of the finitely many norm entries.

Proposition 5.3 (Translations on the reflection Hilbert space)**.** Under Proposition 5.1, fix one of its mirrors and its inward unit normal e. Normal translations by ue, $u \geq 0$, descend to a strongly continuous semigroup $T_u = e^{-uH}$ of positive self-adjoint contractions, with $H \geq 0$. Tangential translations descend to a strongly continuous unitary group $U_y = e^{iy\cdot P}$, $y \perp e$. The generators H and P have a joint spectral measure. Both translation actions fix Ω. ■

Proof. The identity $\Theta_\Pi \tau_{ue} = \tau_{-ue}\Theta_\Pi$ and joint translation invariance give

$$q_{G,\Pi}(F, \tau_{ue}J) = q_{G,\Pi}(\tau_{ue}F, J).$$

We first obtain a contraction on compact separated polynomials. At a fixed cutoff, use a corresponding lattice normal shift v into the chosen half-space. The same reflection identity and semidefinite

Cauchy–Schwarz give

$$\|\tau_v A\|_{\Pi_j}^2 \le \|A\|_{\Pi_j}\,\|\tau_{2v}A\|_{\Pi_j}.$$

The cutoff norm is at most the supremum norm of the bounded cylinder function, uniformly over its translates at this cutoff. If the initial norm is zero, the inequality gives a zero shifted norm. Otherwise iterating with $2^k v$ and letting $k \to \infty$ gives contraction. This argument uses the cutoff supremum bound only before taking the cutoff limit.

Choose v_j in the lattice normal direction converging to ue, with nonnegative normal component. For a diagonal normal one can use integer multiples of $a_j(e_i \pm e_l)$. Pass the two squared norms to the limit by separated moving-test convergence as in the preceding proof. This proves contraction on the compact tensor core. Continuity of the reflected form and the translation on Schwartz tests extend it to finite sequences and preserve the null space. It therefore extends to the completed quotient. The reflection identity gives self-adjointness, and

$$q_{G,\Pi}(F, \tau_{ue}F) = q_{G,\Pi}(\tau_{(u/2)e}F, \tau_{(u/2)e}F)$$

gives positivity. Tangential translations and their inverses preserve the form, so they descend to unitaries. They commute with the normal translations by the translation law.

As the translation tends to zero, its tests converge in Schwartz topology. Expand the squared norm of the difference of two translated classes into form entries. Their continuity gives strong continuity on the test classes, and the norm bounds extend it to the completion. The positive contraction semigroup has a nonnegative self-adjoint generator. In particular T_1 is injective. A vector in its kernel lies in the kernel of every $T_{2^{-k}}$ by self-adjointness and the semigroup law, and strong continuity then gives the zero vector. One may take $H = -\log T_1$ by spectral calculus, using positive roots and continuity to recover all T_u. The tangential translations commute with T_1 and hence with its spectral projections. The joint spectral theorem for the strongly continuous abelian group $(s, y) \mapsto e^{isH}U_y$ gives the asserted spectral measure. The degree-zero class is fixed because it has no translated arguments. □

For the coordinate time plane write $q_G = q_{G,\Pi}$ and $\|F\|_G = q_G(F,F)^{1/2}$. For a compact test h, $\phi(h)$ in a polynomial denotes the entry with test h. Write $G(P)$ for the sum of moments of a finite polynomial P, including its constant term. In the expressions below each moment has separated arguments unless the stated map has already been extended by continuity. Use the source norm in eq. 2.iii.

Proposition 5.4 (Uniform insertion on the actual reflection form)**.** Use the same cutoff states, reference subtraction and separated convergence as in Proposition 5.1. Suppose the ordinary source and zero-source floor bounds and the fixed-layer complete rooted read hypotheses of [3, Theorem C] hold uniformly for those data. Let F, J be compact separated C^1 tensor polynomials with supports in positive time. For $l \le 1$ a power of L, and real $h \in C_c^1$ supported in the radius-$l/4$ ball centered at spatial point $y \in \mathbb{R}^3$ and time zero, one has

$$\left|G\big(\Theta\overline{J}\,\phi(\tau_{(l/2)e_4}h)\tau_{le_4}F\big)\right| \le C\|h\|_\sharp \|J\|_G \|F\|_G. \tag{5.ii}$$

The operation $\phi(\tau_{(l/2)e_4}h)\tau_{le_4}$ descends to a bounded map of norm at most $C\|h\|_\sharp$ on the quotient completion. If $0 < \eta \le 1$, $0 < c < 1/64$, and $k \in C_c^1$ is supported in the radius-$c\eta$ ball about z with $|z_4| \le \eta/8$, then for F, J strictly above time $\eta/4$ one has

$$\left|G\big(\Theta\overline{J}\,\phi(k)F\big)\right| \le C\eta^{-4}\|k\|_\sharp\, \|\tau_{-(\eta/4)e_4}J\|_G \|\tau_{-(\eta/4)e_4}F\|_G. \tag{5.iii}$$

The constants are independent of the degree and number of monomials of the surrounding polynomials, the states, the cutoff sequence and the support positions. The assertions also hold for the self-centered hierarchy G^0 of the same selected state sequence. ■

Proof. Use [3, Proposition 11.2], with its actual states ρ_j, ν_j and uniform source hypotheses. In its lattice insertion the plane at $a_j/2$ approximates time zero, and the central cell has time coordinate $a_j/2 + l/2$. Since l/a_j is odd for large j, the center is a lattice site. Choose its spatial center within $O(a_j)$ of y. Translate all tests by this common small displacement with time component $a_j/2$, before translating the insertion to the middle of its cell. The norm terms agree with the lattice reflections of these shifted arguments, and the passed fixed gaps give exactly the same limiting values as in Proposition 5.1. The two-state estimate in the cited source uses the chosen ν_j in the inserted scalar as well. This verifies that eq. 5.ii applies on the present hierarchy without a new centering or a new subsequence.

The polynomial $\phi(\tau_{(l/2)e_4}h)\tau_{le_4}F$ has strictly positive, internally separated arguments. Use it as J in eq. 5.ii. When its norm is nonzero, divide the squared-norm bound by that norm. This gives the bounded map and sends every null class to zero. When its norm is zero the same norm bound holds without division. For complex tests the estimate follows by splitting real and imaginary parts and adjusting C.

The near-mirror bound is [3, Corollary 11.3], which uses the same limiting forms and the contraction of positive normal translations. Its hypotheses keep the reflected left and right entries at distance at least $\eta/8 - c\eta$ from the insertion support. For clarity, the intermediate slab estimate in that corollary gives

$$\|\phi(f)\tau_{4\eta e_4}F\|_G \le C(1+R)^3\eta^{-4}\|f\|_{C^1}\|F\|_G$$

for tests f supported in $\eta \le x_4 \le 3\eta$ of diameter at most R. It is a norm estimate with an explicit time separation. The near-mirror inequality concerns a sesquilinear form with the stated margins. Neither requires a value for coincident insertions.

For G^0 apply the same insertion result with both cutoff states equal to ρ_j. The self-centered moments converge on the same subsequence by Proposition 3.1, and the hypotheses and constants are uniform for these choices. □

6. Phase means and nontriviality

Proposition 6.1 (Changing the finite phase mean)**.** Suppose the selected full reference-centered hierarchy has $G_1(f) = b\int f$ with $b \in \mathbb{R}$, and G^0_n is its self-centered hierarchy on separated tuples. Use their continuous flat extensions from Proposition 4.1, and suppose the reflection form of Proposition 5.1 is positive for at least one of them. For $A \subset \{1,\dots,n\}$ let I_A integrate the variables indexed by A, retaining the other slots in order. Then on $\mathcal{S}_{\ne,n}$

$$G^0_n(F_n) = \sum_{A\subset\{1,\dots,n\}} (-b)^{|A|} G_{n-|A|}(I_A F_n). \tag{6.i}$$

The inverse formula has b in place of $-b$ and interchanges G with G^0. Both have zeroth moment one. If one hierarchy has a bound $A_0B^n(n!)^p|F_n|_{kn}$ with $p \ge 0$, so does the other, with k replaced by $\max(k,6)$ and the exponential base enlarged by $C|b|$. Their reflection Hilbert spaces are unitarily intertwined by the shift of tests in eq. 6.i, preserving the vacuum and translation actions. Joint Euclidean invariance and the separated cluster property pass between the two hierarchies whenever established on either one. ■

Proof. On a separated tensor, subtract $b\int f$ in each entry and expand the product. This gives eq. 6.i including $A = \{1,\dots,n\}$, whose remaining test is a scalar. Lemma 4.2 preserves the flatness of the remaining variables and gives $|I_A F_n|_r \le C^{|A|}|F_n|_{r+6|A|}$. Continuity and density then identify the expression with the diagonal-flat extension. If $k \ge 6$, take $r = k(n-|A|)$. The order is at most kn. The factorial at degree $n-|A|$ is at most $(n!)^p$. Summing the binomial coefficients bounds the new exponential base by $B + C|b|$, after increasing the prefactor to cover degree zero.

On finite sequences let C_b send each degree-n test to the lower-degree terms $(-b)^{|A|}I_A F_n$ and sum all terms of each resulting degree. The two opposite shifts compose to the identity: for a fixed

set of removed slots the coefficient in the composition is the binomial sum of b and $-b$. The maps preserve positive-side flatness of the retained variables. They also preserve ordered positive-time tests, because any violation of strict order in the retained times violates it in the original tuple. The maps commute with conjugation and affine isometries, since b is real and the integrated Lebesgue measure is invariant. On a product of two test groups, integration over subsets splits into the independent subsets of those groups. It follows that

$$q_{G^0,\Pi}(F,J) = q_{G,\Pi}(C_b F, C_b J).$$

The invertibility of C_b proves positivity of both forms from positivity of either one. The map on quotient classes is therefore an isometry with an inverse on the test domain. Its unitary extension fixes degree zero and intertwines the translations. The same subset identities commute with common Euclidean motions. In a clustering expression, the limit for each pair of remaining groups factors, including an empty group, and the finite subset sums then factor. This proves the invariance and clustering claims. □

Corollary 6.2 (The closed physical shape law)**.** Let G be a separated reference-centered limit along one admitted sequence of integrating and subtracting Wilson states. Assume the running, source and reference hypotheses of [4, Theorem A and Corollaries 6.5 and 7.1], with the finite-reference route or its cofinal alternative, and the same physical normalization as in eq. 2.i. For each fixed sufficiently small $d > 0$, require the comparison to apply at all sufficiently late cutoffs of this sequence to every fixed pair in the following class. The tests f, h are nonzero, nonnegative and compact C^1. Each support lies in a ball of radius d, their mutual distance is in $[d, 6d]$, and each $v = f, h$ satisfies

$$\int v \geq \vartheta_* \|v\|_\infty d^4, \qquad \operatorname{Lip} v \leq \|v\|_\infty/(\vartheta_* d), \qquad \vartheta_* = \frac{1}{12000}.$$

Write $G_2^T(f,h) = G_2(f,h) - G_1(f)G_1(h)$, and let $c_* > 0$ be the uniform inverse-coupling logarithmic coefficient in the cited comparison. Put

$$\mathcal{K}(f,h) = \iint f(x)h(y)|x-y|^{-8}\,dx\,dy, \qquad M_d(f,h) = \frac{9\,\mathcal{K}(f,h)}{\pi^4 c_*^2 [\log(1/d)]^2}.$$

There is a nonnegative function $\epsilon_{\mathrm{run}}(d) \to 0$ such that

$$\frac{1}{2}(1-\epsilon_{\mathrm{run}}(d))^4 \;\leq\; \frac{G_2^T(f,h)}{M_d(f,h)} \;\leq\; \frac{3}{2}(1+\epsilon_{\mathrm{run}}(d))^4. \tag{6.ii}$$

In particular, with $m_v = \int v$ and $M_v = \|v\|_\infty$,

$$\frac{c\vartheta_*^2 M_f M_h}{[\log(1/d)]^2} \;\leq\; G_2^T(f,h) \;\leq \frac{C\min\{M_f M_h, d^{-8} m_f m_h\}}{[\log(1/d)]^2}. \tag{6.iii}$$

The function, constants and small-distance threshold can be chosen uniformly over phases realized on these data when the comparison assumptions have the corresponding all-state uniformity. ■□

Proof. For the late admitted cutoffs at scale d, choose $\epsilon_{\mathrm{run}}(d) < 1$ as a uniform upper bound for

$$|\zeta_K(d) - 1|, \qquad |c_* g_K(d)^2 \log(1/d) - 1|.$$

Such an envelope tends to zero by the two running limits in [4]. Dividing the cutoff law by $M_d(f,h) > 0$ gives the product of the squares of these two running factors and $1 + r_{K,\rho}(f,h)$. The latter lies in $[1/2, 3/2]$. The connected correlation is unchanged by deterministic reference subtraction. At a fixed pair its limit exists because the two-point and both one-point functions converge. Pass the resulting closed inequalities to that limit to obtain eq. 6.ii. The bounds require no convergence of the individual running factors as the cutoff is removed at fixed d.

Shrink the distance threshold so that $\epsilon_{\mathrm{run}}(d) \le 1/2$. All point-pair distances on the two supports lie in $[d, 10d]$. Hence $(10d)^{-8} m_f m_h \le \mathcal{K}(f,h) \le d^{-8} m_f m_h$. Use the mass lower bound in the shape class and $m_v \le (\pi^2/2) d^4 M_v$ to obtain eq. 6.iii. □

Corollary 6.3 (A nonzero scalar excitation)**.** Suppose a selected phase is realized by the convergent separated reference-centered moments of Proposition 3.1, with the flat extension and positive reflection form of Propositions 4.1 and 5.1. Let its realizing states be admitted periodic Gibbs states at the same bare sequence and physical source normalization as in the two-point comparison of [4, Theorem A and Corollaries 6.5 and 7.1]. Require that at each fixed sufficiently small d the chosen comparison route applies for all sufficiently late cutoffs at that d, with common constants. The selected limit can retain a real mean $G_1(f) = b \int f$. For the fixed-shape real unit-mass tests f_d below, the connected covariance satisfies

$$c\,\frac{d^{-8}}{[\log(1/d)]^2} \le G_2(\theta f_d, f_d) - G_1(\theta f_d) G_1(f_d) \le C\,\frac{d^{-8}}{[\log(1/d)]^2}. \tag{6.iv}$$

Here θ reflects in $x_4 = 0$ and $c, C > 0$ are independent of sufficiently small d. The reflection Hilbert space has a nonzero vector orthogonal to the vacuum already at degree one. ■□

Proof. Let $\varphi(x) = (12/\pi^2)(1 - |x|^2)_+^2$ and $f_d(x) = d^{-4}\varphi((x - 2de_4)/d)$. Radial integration gives unit mass. The profile has amplitude $12/\pi^2$ and Lipschitz constant at most $8/(3\sqrt{3})$ times that amplitude. The rescaled positive and negative supports have radius d and gap $2d$. Use $2d$ as the comparison scale. The supporting balls have radius at most $2d$, and the mass and Lipschitz bounds of that shape class hold with parameter $1/12000$. The amplitude product is $(12/\pi^2)^2 d^{-8}$, and $\log(1/(2d))$ is comparable to $\log(1/d)$ as d decreases. Apply the closed bounds eq. 6.iii from Corollary 6.2 at scale $2d$. They give eq. 6.iv. That corollary takes the cutoff limit at each fixed pair, so this application removes the cutoff before sending d to zero.

In the reflection space, the squared norm of $[\phi(f_d)]_G - G_1(f_d)\Omega$ equals the middle term of eq. 6.iv. The vector is orthogonal to Ω, because $q_G(1, \phi(f_d)) = G_1(f_d)$. It has positive norm and belongs to the completed ordered positive-time sector. To see the latter, approximate the degree-one test in C^1 by smooth tests in a fixed compact neighbourhood strictly above time zero. Their classes converge by the separated moment bound. In a scalar reconstruction that matches these moments it excludes a field proportional to the identity. □

7. Translation-ergodic limits

7.1. **The phase law at one spacing.** Let $\mathcal{I}_\mu$ denote the completed translation-invariant σ-field of a translation-invariant probability measure μ. An ergodic measure is one for which $\mathcal{I}_\mu$ is trivial. Write $\mathcal{E}_{a,\beta}$ for the ergodic members of $\mathcal{G}_{a,\beta}$.

We use the following external lattice estimate [5, uniform plateau estimate, Appendix A]. Its hypotheses are translation invariance and reflection positivity on the closed half-lattice for every coordinate site, coordinate link and diagonal lattice mirror. Let A, B be bounded Borel observables of bonds with both ends in $a[-r, r]^4$, for an integer $r \ge 1$. For $v \in a\mathbb{Z}^4$ and $n = |v|_\infty/a - 2r \ge 1$, it gives

$$\begin{aligned} &\Big|\mu(\overline{A}\,\tau_v B) - \mu\Big(\overline{E_\mu[A \mid \mathcal{I}_\mu]}\,E_\mu[B \mid \mathcal{I}_\mu]\Big)\Big| \\ &\qquad \le C\big[(r/n)^2 + (r/n)^3\big]\, N_{\mu,r}(A)^{1/2} N_{\mu,r}(B)^{1/2}, \\ &N_{\mu,r}(A) = \max_{\lambda,\gamma} \mu\Big(\overline{\theta_\lambda \gamma A}\,\tau_{2rae_\lambda}\gamma A\Big). \end{aligned} \tag{7.i}$$

Here θ_λ reflects in $x_\lambda = 0$, γ ranges over the coordinate sign changes, and τ_v translates supports by v. The terms in the maximum are squared reflection norms at $x_\lambda = ra$. The constant is numerical. The lattice theorem in [1, Theorem B] supplies all the reflection hypotheses for every member of $\mathcal{G}_{a,\beta}$.

Proposition 7.1 (Ergodic Gibbs states)**.** Fix $a > 0$ and $\beta \geq 0$. Assume the uniform lattice plateau estimate eq. 7.i for every member of $\mathcal{G}_{a,\beta}$. The set $\mathcal{E}_{a,\beta}$ is compact. It equals the set of extreme points of $\mathcal{G}_{a,\beta}$. Each $\mu \in \mathcal{G}_{a,\beta}$ is the barycentre of a unique probability law D_μ on $\mathcal{E}_{a,\beta}$. For every bounded Borel F, the evaluation $\rho(F)$ under D_μ has the law of $E_\mu[F \mid \mathcal{I}_\mu]$. The identification also holds jointly for any finite list of such functions. ■

Proof. Let γ_D be the Wilson conditional kernel on a finite bond set D. If h is a bounded invariant density, then $\mu(hF) = \mu(h\gamma_D F)$. In fact, approximate h in $L^1(\mu)$ by bounded continuous cylinder functions. Translate each approximant until its bonds avoid D. Invariance of h and μ preserves its approximation error. The translated approximant can be taken outside the conditional integral defining γ_D. The claimed identity follows in the L^1 limit. In particular, conditioning on any invariant event of positive probability preserves the Gibbs equations.

On the compact metric configuration space, a conditional probability $\omega \mapsto \mu^\omega$ given $\mathcal{I}_\mu$ can be constructed by taking conditional expectations on a countable uniformly dense rational vector space of real continuous functions containing 1. On one set of full measure they are linear, positive on the nonnegative members, have absolute value at most $\|f\|_\infty$ on a test function f, and take value 1 on the constant 1. Their continuous extensions are positive by uniform approximation and the constant function. The Riesz representation theorem gives probability measures. Approximation gives measurability on continuous functions, and the bounded monotone class theorem gives the conditional identity on bounded Borel functions. Use countable continuous cylinder determining families in the conditional identities for γ_D and the lattice translations. The invariant-density argument and translation invariance show that $\mu^\omega \in \mathcal{G}_{a,\beta}$ almost surely.

For a real continuous local function A, apply eq. 7.i to μ and to each μ^ω. Dominated convergence in the disintegration of $\mu(A\tau_v A)$ shows that

$$\int \left[\mu^\omega(E_{\mu^\omega}[A \mid \mathcal{I}_{\mu^\omega}]^2) - \mu^\omega(A)^2\right] d\mu(\omega) = 0.$$

The two integrands are measurable because the first is the limit of local correlations. The integrand in brackets is nonnegative. A countable dense family of real continuous local functions is dense also in L^2 for each component. The vanishing conditional variances on this family prove ergodicity almost surely. Push the conditional law to $\mathcal{E}_{a,\beta}$ to obtain D_μ.

A nontrivial invariant event splits its measure into two different invariant Gibbs states by the density argument. Conversely a constituent of a nontrivial convex decomposition has a bounded invariant Radon–Nikodym density with respect to the barycentre. An ergodic barycentre makes that density constant. This proves the extreme point characterization.

For each fixed continuous local A, B, the right side of eq. 7.i has a state-independent bound using their supremum norms. Hence a weak limit of ergodic Gibbs states factorizes these two correlations at large separation. The plateau identity and the same L^2 argument show that the weak limit is ergodic. This proves compactness. Finally, a law on $\mathcal{E}_{a,\beta}$ with barycentre μ has all its mixed evaluation moments determined by μ. To obtain the product of q local evaluations, translate the corresponding observables far apart, taking successive limits with the other translates fixed. The uniform plateau bound permits each limit through the law. Products of continuous local evaluations form an algebra that separates the compact set $\mathcal{E}_{a,\beta}$. Stone–Weierstrass gives uniqueness of the law. □

7.2. **A common cutoff sequence for the continuum phases.** Let X be a separated tuple of real compact C^1 tests, or a finite linear combination of such tuples. The empty tuple is the unit. Write X_j for its lattice polynomial formed with $X_j^\nu(f)$ from eq. 2.i. Products of tuples below are used when the resulting supports are pairwise disjoint. Assume that the reference states ν_j define a separated limit G, so that $\nu_j(X_j) \to G(X)$ for every such X. This is one choice of the integrating sequence in the separated compactness proposition.

Theorem A (The phase law of a reference limit)**.** Let the reference states $\nu_j \in \mathcal{G}_{a_j,\beta_j}$ converge on the reference-centered separated compact moments to G along one cutoff sequence, and assume the all-state bound eq. 2.iv at these data. Use the lattice plateau estimate eq. 7.i. There is a compact metrizable space $\mathcal{P}$ with a probability law λ, with $\operatorname{supp}\lambda = \mathcal{P}$, whose points are distinct separated hierarchies G_ξ. For each $\xi \in \mathcal{P}$ there are states $\rho_{j,\xi} \in \mathcal{E}_{a_j,\beta_j}$ along the given cutoff sequence such that

$$G_\xi(X) = \lim_j \rho_{j,\xi}(X_j)$$

for every separated compact tuple. Each G_ξ satisfies the same graded estimate as G. The evaluations on compact tuples are continuous in ξ, and

$$G(X) = \int_{\mathcal{P}} G_\xi(X)\, d\lambda(\xi). \tag{7.ii}$$

For compact tuple polynomials X, Y and all sufficiently large $|v|_\infty$,

$$|G_\xi(X\,\tau_v Y) - G_\xi(X)G_\xi(Y)| \le C_{X,Y}|v|_\infty^{-2}. \tag{7.iii}$$

The constant and threshold are independent of ξ. The phase law is determined by G through the formula

$$\int_{\mathcal{P}} \prod_{i=1}^{q} G_\xi(X^i)\, d\lambda(\xi) = \lim_{|v_q|\to\infty} \cdots \lim_{|v_2|\to\infty} G(X^1\,\tau_{v_2}X^2 \cdots \tau_{v_q}X^q), \tag{7.iv}$$

where each inner limit has the outer translates fixed and where the other, fixed groups are already separated. Any compact Hausdorff representation with continuous separating evaluations, a regular law of full support, and these mixed moments is related to $(\mathcal{P}, \lambda)$ by an evaluation-preserving measure-preserving homeomorphism. ■□

Proof. For any fixed X, Y, put their support sets in a physical cube $[-R, R]^4$ with R a sufficiently large integer and a positive margin to all its faces. For $a_j \le 1$ put $r_j = R/a_j$. Each lattice polynomial reads only bonds in $a_j[-r_j, r_j]^4$. The reflected pair in $N_{\rho,r_j}(X_j)$ is a linear combination of products of separated compact tests with a fixed positive gap at the common face. Its norm is bounded uniformly in j, ρ by eq. 2.iv. The same holds for Y. Sign changes, mirrors and lattice translations carry $X_j^\nu(f)$ to the corresponding transformed test since ν_j is invariant under these transformations. For an ergodic ρ, eq. 7.i now gives

$$|\rho(X_j\,\tau_v Y_j) - \rho(X_j)\rho(Y_j)| \le C_{X,Y}|v|_\infty^{-2} \qquad (v \in a_j\mathbb{Z}^4,\ |v|_\infty \ge 4R+1). \tag{7.v}$$

Complex polynomials follow by using $\overline{X}$ in the plateau bound or by linearity.

Choose a countable real determining family of tuples. One can use the compact supports of finite unions of rational closed boxes with interiors, and a countable dense set of C^1 tests supported in each such union. Each separated tuple is approximated in C^1 in fixed separated unions. This follows by first covering its compact supports in separated open neighbourhoods by finite box unions. The graded bound and multilinearity give equicontinuity on these fixed containers, by telescoping in the entries. Consequently convergence on the countable family implies convergence on every separated tuple for these cutoff laws.

The evaluations on the determining family take values in a countable product $\mathcal{K}$ of compact real intervals, by the graded bound. For $\rho \in \mathcal{E}_{a_j,\beta_j}$ let $e_j(\rho) \in \mathcal{K}$ be its vector of evaluations. Let $\lambda_j = (e_j)_* D_{\nu_j}$. We show that the mixed coordinate moments of λ_j are Cauchy. Given any finite product of evaluations, first approximate the product of its first two factors by the evaluation of a single tuple with one group translated by a fixed large integer vector, using eq. 7.v. Proceed with the third group and then each remaining group. Each new translation is chosen after the existing groups are fixed. All supports can therefore be kept separated, all choices work uniformly in the state and cutoff, and the total error can be made arbitrarily small. The error terms are multiplied only by uniformly bounded evaluations of the remaining factors. The resulting single evaluation, integrated against D_{ν_j}, converges by the assumed convergence of ν_j. This proves the

Cauchy assertion. Since the coordinate algebra separates $\mathcal{K}$, compactness and Stone–Weierstrass give a unique weak limit λ.

Put $\mathcal{P} = \operatorname{supp}\lambda$. For $\xi \in \mathcal{P}$, every open neighbourhood of ξ has positive λ_j-mass for all sufficiently large j. For a decreasing sequence of such neighbourhoods of radius tending to zero, choose states in their inverse images for all sufficiently large indices, increasing the neighbourhood index along the full tail. This yields $e_j(\rho_{j,\xi}) \to \xi$. Uniform equicontinuity in the test functions extends the limits to all separated tuples. It also proves continuity of the resulting evaluations in ξ, by uniform approximation by the coordinate evaluations. The bound and eq. 7.ii follow by passage to the limit, first on the countable family and then by approximation. The translation and signed coordinate symmetries pass on each realizing sequence as in Proposition 3.1.

Pass eq. 7.v to the limit for translations in $\bigcup_{k\ge 0} L^{-k}\mathbb{Z}^4$. These are eventually lattice translations and form a dense subgroup. Uniform continuity of the translated tests in C^1, with the separated containers enlarged slightly, extends the estimate to every large real v. This proves eq. 7.iii. To prove eq. 7.iv, apply eq. 7.ii and take the inner translation to infinity. The other groups have compact support and are kept fixed. Their product can be taken as the fixed factor in eq. 7.iii. The bound is uniform in ξ. Repeat with the other translations in the indicated order. For uniqueness, embed both compact phase spaces in the product of real evaluation coordinates indexed by all separated real tuples. Each map is an embedding because its coordinates are continuous and separate points. On the compact union of their images, the two laws agree on the polynomial coordinate algebra by eq. 7.iv. Stone–Weierstrass gives equality of the regular laws. Full support gives equality of the images, and the two embeddings give the claimed homeomorphism. □

Proposition 7.2 (The mean of a phase)**.** There is a real continuous function b on $\mathcal{P}$ such that $G_{\xi,1}(f) = b(\xi)\int f$ and $\int_{\mathcal{P}} b\,d\lambda = 0$. For each selected phase the hierarchies G_ξ and G^0_ξ, obtained from one another by adding or subtracting the mean, are limits along the same realizing sequence. More precisely G^0_ξ uses $\rho_{j,\xi}$ for the subtraction in eq. 2.i, and

$$G_{\xi,n}(f_1,\dots,f_n) = \sum_{I\subset\{1,\dots,n\}} G^0_{\xi,|I|}(f_I)\prod_{i\notin I} b(\xi)\int f_i. \tag{7.vi}$$

For a fixed lattice plaquette p_j at each spacing,

$$\int b^2\,d\lambda = \lim_j 36a_j^{-8}\operatorname{Var}_{\nu_j}(E_{\nu_j}[A_{p_j}\mid \mathcal{I}_{\nu_j}]). \tag{7.vii}$$

In particular, for separated f, h the mixture two-point function is the average of $G^0_{\xi,2}(f,h)$ plus $(\int b^2\,d\lambda)\int f\int h$. ■

Proof. Translations and signed coordinate permutations make the plaquette mean the same at every plaquette of the same spacing. Put $b_j(\rho) = 6a_j^{-4}[\rho(A_{p_j}) - \nu_j(A_{p_j})]$ and $R_j(f) = a_j^4\sum_p f(x_p)/6$. Then $\rho(X_j^\nu(f)) = b_j(\rho)R_j(f)$. The six shifted lattice Riemann sums give $R_j(f)\to\int f$. Use a real test f_* of integral one in the determining family. Its evaluations and the graded bound show that b_j is bounded uniformly for large j, and on a realizing sequence $b_j(\rho_{j,\xi}) \to G_{\xi,1}(f_*)$. This proves the singleton formula and continuity. Equation 7.ii gives the zero average. Expand the finite products after the change of subtraction to obtain eq. 7.vi.

The conditional identification in Proposition 7.1 gives $\int b_j(\rho)^2\,dD_{\nu_j}(\rho)$ equal to the variance expression at spacing a_j. Since $R_j(f_*)\to 1$, weak convergence of the phase laws on this bounded coordinate gives eq. 7.vii. The last assertion follows from eq. 7.vi at order two. The finite-spacing plateau formula also identifies the variance before rescaling as the limit of the connected plaquette correlation under far lattice translation at that spacing. □

8. Euclidean invariance of the limiting moments

Proposition 8.1 (The joint transfer spectrum)**.** For a limit through the infinite-lattice states of Proposition 5.1, use the reflection Hilbert space of any coordinate mirror. At each cutoff assume the external lattice carrier of [5, Uniform lattice plateau estimate and the Euclidean Hilbert bridge]: for joint one-step spectral values $e^{-\epsilon} \in (0,1]$ in the normal direction and e^{ip}, $p \in [-\pi,\pi]$, in any tangential coordinate, $\sinh(\epsilon/2) \geq |\sin(p/2)|$. Zero normal-transfer values are also permitted. The nonnegative normal generator H and the tangential generators P_1, P_2, P_3 have joint spectral support in

$$E \geq \max_i |p_i|. \tag{8.i}$$

■

Proof. Use the site mirror at zero and let $T_j, V_{j,i}$ be the normal transfer and the one-step tangential translations at cutoff a_j. The coordinate and diagonal reflection hypotheses of the assumed carrier are supplied by [1, Theorem B]. Apply it on the joint spectral variables $T_j = e^{-\epsilon} > 0$ and $V_{j,i} = e^{ip}$. Convexity of sinh on the nonnegative axis and $|\sin(kx)| \leq k|\sin x|$ give, for every positive integer k,

$$(1 - T_j^k)^2 \;\geq\; T_j^k(2 - V_{j,i}^k - V_{j,i}^{-k}).$$

The inequality includes the kernel of T_j, where its two sides are 1 and 0. Fix $t > 0$ and choose $k_j a_j \to t$. On a compact positive-time polynomial each term is a reflected moment with at most a bounded positive time shift and a bounded tangential shift. The reflected groups retain their fixed separation. The centered lattice observables intertwine these lattice shifts and the site reflection. The graded bound and moving-test convergence pass the inequality to

$$(1 - e^{-tH})^2 \;\geq\; e^{-tH}(2 - e^{itP_i} - e^{-itP_i})$$

on the dense compact positive-time classes and hence on the whole Hilbert space. Proposition 5.3 supplies the strongly commuting generators and $H \geq 0$. For the countable sequence $t = 1/m$ the joint multiplier $2e^{-tE}[\cosh(tE) + \cos(tp_i) - 2]$ is nonnegative almost everywhere. Dividing the bracket by t^2 and taking $m \to \infty$ proves $E^2 \geq p_i^2$ on a common set of full spectral measure. □

8.1. **Analytic criteria for rotations.** We use four external criteria from [5, analytic insertion densities, planar angular kernels, two-point angular elimination, and all-orders angular closure]. Their common input is a centered scalar hierarchy on separated compact C^1 tuples. The hierarchy must be symmetric, real, translation invariant, invariant under signed coordinate permutations, and continuous on each separated support collection with position-uniform bounds at fixed diameters and distances. Its coordinate reflection forms must have the vacuum vector and the translation actions of Proposition 5.3 with spectrum eq. 8.i. The criteria use the following further data and implications.

(1) The insertion-to-density result uses eq. 5.iii with a single exponent p in place of 4, where $p \geq 0$, uniformly over all compact outside polynomials. Fix $0 < c < 1/64$ and $0 < \eta \leq 1$. For any coordinate plane, write $y = (x_\lambda, x_\mu)$ with x_λ normal time and z for the remaining coordinates. For a real compact C^1 transverse test $\varphi(z)$, the insertion $\alpha(y)\varphi(z)$, with α compact C^1 and supported in $|x_\lambda| < c\eta$, has the bound

$$C_\varphi \eta^{-(p+5)} \|\alpha\|_{L^1} \|F\|_{(\eta/4)} \|J\|_{(\eta/4)}$$

when the outside supports are strictly above $\eta/4$. Here the pulled-down norm is the reflection norm after translation by $-(\eta/4)e_\lambda$. The result supplies compatible analytic operator densities between positive time margins and translation identities for their regularized point-plate operators.

(2) The planar kernel criterion uses these L^1 plate bounds, the indicated spectral carrier in the plane, the common local continuity and the signed coordinate symmetries. It supplies holomorphic product kernels and continued reflection Gram matrices. Their densities agree with the compact separated moment functional on ordinary planar ordering charts. In particular for all distinct planar pairs there is a continuous two-point density $K_{\varphi,\psi}$ with

$$G_2^0(\alpha\otimes\varphi,\beta\otimes\psi)=\iint \alpha(y)\beta(y')K_{\varphi,\psi}(y'-y)\,dy\,dy'$$

when the planar profiles have disjoint supports. For $t>2s>0$ this density in the coordinate frame has the representation

$$K_{\varphi,\psi}(t,x)=\langle z_{\varphi,s},e^{-(t-2s)H+ixP}z_{\psi,s}\rangle,\qquad z_{\varphi,s'}=e^{-(s'-s)H}z_{\varphi,s}\quad(s'>s).$$

At order n the angular coefficients are real analytic away from planar coincidences and the band contains only $|m|\le Cn$ with characters e^{4imu} for planar rotation angle u. The continued Gram identities give the highest-coefficient positivity on reflected half-plane configurations and the inequality $2b_{k+l}\le b_{2k}+b_{2l}$ for the global largest absolute angular indices b_j. These statements are conclusions of the kernel construction.

(3) The order-two criterion uses the continuous two-point density and the coordinate spectral representation just stated. It also requires positivity at the diagonal mirror whose normal makes angle $\pi/4$ with the time axis in that plane. Its short-distance input is

$$0\le G_2^0(f,h)\le u(d)\|f\|_\infty\|h\|_\infty,\qquad u(d)\to 0,$$

uniformly on the shape class of Corollary 6.2, at its fixed parameter $\vartheta_*=1/12000$, for every sufficiently small $d>0$. It gives radiality for each equal-label planar kernel and then for mixed labels by polarization. Thus $b_2=0$.

(4) The all-orders criterion uses the band, Gram identities, top-coefficient positivity, the coordinate reflection and cone hypotheses, and $b_2=0$. It also uses the off-diagonal distribution supplied by Proposition 4.1. Its polygon and integer degree argument removes the remaining harmonics. Its common-cut argument identifies the invariant moment functional on arbitrary separated compact four-dimensional tuples, including when planar projections overlap. It yields invariance under each coordinate-plane rotation, and hence under O(4) with the signed coordinate symmetries.

Theorem B (Rotations of the realized limits)**.** Let G be a separated limit through the stated infinite-lattice Wilson states, with the uniform ordinary-source and insertion assumptions of the setting. Suppose its own realizing sequence satisfies the assumptions of the two-point shape bound in [4, Theorem A and the ultraviolet shape corollary], using that theorem or its cofinal comparison alternative at the same bare data. Use the external lattice carrier of proposition 8.1 and the four analytic criteria stated in section 8.1. Then G^0 and G are invariant under the Euclidean group on separated tuples and on $\mathcal{S}_{\neq,n}$ at each order. When the comparison assumptions are uniform over all states at the fixed data, this assertion applies to every phase in Theorem A and to its barycentre. ■□

Proof. The self-centered hierarchy G^0 is realized along the same cutoff sequence as G, with the integrating states also used as the subtracting states. Proposition 3.1 and eq. 2.iv give the scalar symmetries and the required position-uniform C^1 continuity. Propositions 5.1 and 5.3 give the coordinate and diagonal positivity, the Hilbert translation actions and the unit vacuum. Proposition 8.1 gives the carrier in every coordinate picture. Proposition 5.4 on G^0 gives the insertion hypothesis with $p=4$. Consequently the first two criteria supply the plate bound with $p+5=9$, the planar density and its spectral representation, and the all-orders band and Gram data.

Additive mean shifts leave connected two-point functions unchanged. Corollary 6.2 gives the closed positive-shape law on the specified class with its exact ϑ_*. Its upper bound can be taken

to be $u(d) = C[\log(1/d)]^{-2}$ for small d. The sign follows from the lower bound on that class. The diagonal mirror required by the order-two criterion is one of the limiting diagonal lattice reflections. Thus the third criterion gives $b_2 = 0$, and the fourth gives rotation invariance on all separated compact tuples. Translations were already available. Restore $b \int f$ by the finite subset expansion, which commutes with every isometry. The uniqueness and density of the flat Schwartz extension in Proposition 4.1 pass each invariance identity to the entire flat space. In the phase-law application, every step uses the states $\rho_{j,\xi}$ from Theorem A, or the reference states themselves for the barycentre. □

Corollary 8.2 (Isometries of the canonical phase law)**.** Under Theorem A, every Euclidean isometry of the mixture G induces a measure-preserving homeomorphism of $\mathcal{P}$ acting by transport on the evaluations. In particular, if G is O(4)-invariant then each G_ξ is O(4)-invariant. ■□

Proof. Let r be an isometry of G and let R be its linear part. Give the same phase space the transformed evaluations $X \mapsto G_\xi(r^{-1}X)$. They are continuous and separate points. Since $r^{-1}\tau_v = \tau_{R^{-1}v}r^{-1}$, the mixed moment formula eq. 7.iv and invariance of G identify all their mixed moments with those of the original evaluations. The uniqueness assertion of Theorem A gives a measure-preserving homeomorphism carrying one system of evaluations to the other. Separation of points gives the group law for these homeomorphisms.

Each phase is invariant under the coordinate reflection θ_1 by its lattice realization. If G is rotation invariant, every rotate of G_ξ is another phase by the first assertion, and hence is invariant under θ_1. This says that G_ξ is invariant under every conjugate of θ_1 by an orthogonal map. These hyperplane reflections generate O(4). The same identities hold on the flat Schwartz extension by density. □

Proposition 8.3 (The zero-energy space and spatial rotations)**.** Use the realized phase law of Theorem A and the time reflection spaces of Propositions 5.1 and 5.3. Let H be the normal generator in the reference mixture space and P_0 its projection onto $\ker H$. On compact separated positive-time polynomials the map

$$P_0[F]_G \longmapsto (\xi \longmapsto G_\xi(F))$$

extends to a unitary $\ker H \to L^2(\mathcal{P}, \lambda)$ sending Ω to 1. If $\mathcal{P}$ has $m < \infty$ points, then $\dim \ker H = m$. If $\mathcal{P}$ is infinite, its zero-energy space has countably infinite Hilbert dimension. In each individual phase the kernel of the normal generator is the vacuum line. Its spatial translation matrix elements tend to the vacuum product for arbitrary reflection Hilbert vectors.

For any of these hierarchies satisfying Theorem B, spatial O(3) acts by a strongly continuous unitary representation on its reflection space. It fixes the vacuum, commutes with the normal generator and rotates the spatial translations. The joint spectral carrier in eq. 8.i then improves to $E \geq |p|$. ■

Proof. By spectral calculus $T_t = e^{-tH}$ converges strongly to P_0 as $t \to \infty$. For compact positive-time polynomials F, J, the barycentre formula and the uniform phase cluster bound give

$$\begin{aligned}\langle [F]_G, T_t[J]_G \rangle &= \int_{\mathcal{P}} G_\xi(\Theta\overline{F}\,\tau_{te_4}J)\,d\lambda(\xi) \\ &\longrightarrow \int_{\mathcal{P}} \overline{G_\xi(F)}\,G_\xi(J)\,d\lambda(\xi).\end{aligned}$$

Each reflected tuple is separated. The identity $G_\xi(\Theta\overline{F}) = \overline{G_\xi(F)}$ uses reality and reflection invariance of the phase. Taking the spectral limit proves the asserted isometry on the projection of the dense compact core.

The evaluation functions on that core contain constants, are closed under conjugation and separate $\mathcal{P}$. For separation, translate any compact separating tuple far enough into positive time and

use phase translation invariance. Products of evaluations are in their uniform closure because

$$G_\xi(F)\,G_\xi(J) = \lim_{t\to\infty} G_\xi(F\,\tau_{te_4}J)$$

uniformly in ξ, by the same cluster bound. For large t the product on the right has separated positive-time arguments. The uniform closure is therefore a conjugation-stable subalgebra of $C(\mathcal{P})$. Stone–Weierstrass gives density in $C(\mathcal{P})$, and hence in $L^2(\mathcal{P},\lambda)$. The isometry extends onto this space. If $\mathcal{P}$ is finite, each point has positive mass by full support. If it is infinite, any finite number of distinct points have disjoint open neighbourhoods of positive measure. Their indicators give arbitrarily large orthogonal sets. Since L^2 of a compact metrizable probability space is separable, its Hilbert dimension is countably infinite.

For an individual phase the same normal cluster limit gives the rank-one vacuum projection on the compact core, and hence $\ker H = \mathbb{C}\Omega$. With spatial translations the compact cluster bound gives the same matrix-element limit. Approximate each of two arbitrary vectors by compact classes. Unitarity bounds the two approximation errors independently of the spatial shift, so the limit passes to the entire space.

For $R \in \mathrm{O}(3)$, let the pullback act by $F((R^{-1}\mathbf{x}_1,t_1),\dots,(R^{-1}\mathbf{x}_n,t_n))$. Under Euclidean invariance this action preserves the positive-time flat domain and commutes with time reflection. It is an isometry for the reflection form, with inverse given by the inverse rotation. Thus it descends to a unitary V_R. Spatial signed permutations already give these unitaries under the coordinate symmetries alone. Pullback is continuous in the Schwartz topology as R varies. The form then gives strong continuity on finite test sequences, and the unitary norm gives it on the completion. The exact actions on tests show $V_R T_t = T_t V_R$ and $V_R U_y V_R^{-1} = U_{Ry}$, and $V_R\Omega = \Omega$. The joint spectral measure therefore transforms by spatial rotation of its momentum variables. The cube inequality $E \geq |(Rp)_i|$ holds for each rotation and coordinate. Intersect its supports over a countable dense set of rotations, and use continuity in R to obtain $E \geq |p|$. □

The vacuum-line conclusion does not give a positive gap above zero. For example, take $H = 0 \oplus M_E$ on $\mathbb{C} \oplus L^2((0,\infty), Ee^{-E}\,dE)$, where M_E is multiplication by E. Its kernel is the first summand, but the spectrum on the second reaches zero. The vector $v = 0 \oplus 1$ has $\langle v, e^{-tH}v\rangle = \int_0^\infty Ee^{-(1+t)E}\,dE = (1+t)^{-2}$. This polynomial decay in time coexists with spectral support arbitrarily near zero.

9. The corrected scalar reconstruction theorem

For $n \geq 1$ write $\mathcal{S}_{\mathrm{ord},n}$ for the Schwartz functions all of whose derivatives vanish outside

$$\{(x_1,\dots,x_n) : 0 < (x_1)_4 < \cdots < (x_n)_4\}.$$

Set $\mathcal{S}_{\mathrm{ord},0} = \mathbb{C}$. This space is a subspace of $\mathcal{S}_{+,n}$. The ordered positive-time quotient is the Hilbert completion for the restriction of the reflection form in eq. 5.i to finite ordered sequences.

The growth condition below has content beyond the existence of all moments. For a standard Gaussian Z, the variable $X = e^Z$ has $\mathbb{E}X^n = e^{n^2/2}$. For fixed $A,B,p > 0$, the logarithm of $AB^n(n!)^p$ grows only as $O(n\log n)$. Thus finiteness at every degree does not supply the uniform factorial growth required for reconstruction.

Proposition 9.1 (Applicability of scalar reconstruction)**.** Let the selected scalar hierarchy G_n have the diagonal-flat extension of Proposition 4.1, a positive full reflection form as in eq. 5.i, scalar permutation symmetry, and full Euclidean invariance. Suppose its reflected compact separated tensor polynomials cluster under all large spatial translations in nonzero directions, with the product of the two means as their limit. Assume there are $A, B > 0$, $p \geq 0$, and a nonnegative integer k, independent of n, such that

$$|G_n(F)| \leq AB^n(n!)^p|F|_{kn} \qquad (F \in \mathcal{S}_{\neq,n},\ n \geq 1), \qquad G_0 = 1.$$

Then the corrected Osterwalder–Schrader theorem applies to these scalar moments. It gives a Hermitian scalar Wightman theory, unique up to unitary equivalence in its cyclic realization on a separable Hilbert space, with the Wightman covariance, positivity, spectrum, locality and vacuum cluster properties and a unique translation-invariant vacuum up to scalar multiples. Its Euclidean Green functions agree with G_n on $\mathcal{S}_{\neq,n}$. If the two-point hypotheses of Corollary 6.3 hold, its scalar field is nontrivial. The construction applies with the reference mean or with the self-centered hierarchy in Proposition 6.1. If $G_1(f) = b\int f$ as in Proposition 6.1, write Φ for the reference-centered reconstructed field. The self-centered field on its cyclic polynomial domain is

$$\Phi^0(f) = \Phi(f) - b\left(\int_{\mathbb{R}^4} f(x)\,dx\right) I, \qquad f \in \mathcal{S}(\mathbb{R}^4).$$

The domain generated by the shifted field and the vacuum is the same cyclic polynomial domain. ■

Proof. The corrected theorem is $E' \to R'$ in [9, Section IV.1, pp. 287–288, equation (4.1)]. The Euclidean space in its linear growth condition is the Schwartz subspace flat on every coincidence diagonal, defined in [9, Section II, p. 284]. This is precisely $\mathcal{S}_{\neq,n}$. The radial weight and derivative cutoff in its Section II, equation (2.1), use the convention of eq. 2.v. Choose an integer $s \geq \max(1, k)$ and put $B_+ = \max(1, B)$. Since $n! \geq 2^{n-1}$ for $n \geq 1$, the sequence $AB_+(n!)^{p+\lceil \log_2 B_+ \rceil}$ is a factorial majorant and

$$|G_n(F)| \leq AB_+(n!)^{p+\lceil \log_2 B_+ \rceil}|F|_{sn}.$$

Thus the linear order and factorial coefficient are uniform in the degree. If one instead uses the total-monomial seminorm

$$q_{j,n}(F) = \max_{|\alpha|+|\gamma|\leq j} \sup_{x\in\mathbb{R}^{4n}} |x^\gamma \partial^\alpha F(x)|,$$

the same verification is quantitative. Expansion of $(1 + \sum_i |x_i|)^r$ bounds $|F|_r$ by $(1 + 4n)^r q_{2r,n}(F)$. At $r = sn$, use $n^n \leq e^n n!$ to bound the extra factor by $((5e)^s)^n (n!)^s$. This again spends only a linear order and a factorial coefficient. The growth hypothesis is $E0'$.

For $E1$ and $E3$ use Euclidean invariance and scalar permutation symmetry on the flat spaces. The positivity test of [8, Sections 2–3, pp. 86–88] uses the ordered positive-time sequences. Its reflection conjugates and reverses the first group of variables. Permutation symmetry removes the reversal inside each G_{m+n}, so the required form is the restriction of eq. 5.i. This gives $E2$. The Hermiticity of the form, paired with degree zero, gives $G_n(\Theta\overline{F}) = \overline{G_n(F)}$ on positive-time tests. Euclidean reflection invariance removes Θ. Translate each compact separated tensor into positive time and then use flat-test density. This proves the scalar reality identity on the full diagonal-flat domain. The ordered quotient embeds isometrically in $\mathcal{H}_{G,\{x_4=0\}}$. Its closed range contains the vacuum. The spatial translations preserve strict time order and act unitarily as shown below, so they preserve this closed range. This comparison uses the ordered domain directly.

For $E4$ take any two finite ordered Schwartz sequences. Their classes in the full reflection space can be approximated by compact separated positive-time tensor polynomials, by Propositions 4.1 and 5.1. Spatial translations preserve the reflection form by translation invariance and have inverse translations. They therefore act unitarily on the quotient and its completion, as in Proposition 5.3. Replacing either vector by its approximation changes any translated matrix coefficient by at most the norm of the error times the norm of the other vector, uniformly in the translation. The same

bound holds for the vacuum coefficients. The compact reflected cluster law therefore passes to all vectors of the full reflection space, in particular to the ordered sequences. This extension is qualitative; it asserts no common decay rate for arbitrary vectors. This is the spatially translated cluster condition in [8, Section 3, p. 88]. Thus the corrected theorem applies with $E0'$ and $E1$–$E4$ to the same selected hierarchy. In its cyclic representation the Wightman cluster property gives the weak spatial-translation limit onto the vacuum line, first on cyclic polynomial vectors and then on all vectors by unitarity. A translation-invariant vector lies in that line. The Wightman polynomial test maps are continuous on each finite direct sum of Schwartz spaces. Their countable dense subsets have dense image in the cyclic Hilbert space, so that space is separable as well.

The scalar connected covariance in eq. 6.iv is unchanged by adding a constant multiple of the identity to a field. A scalar field proportional to the identity would give zero for it. The mean shift of Proposition 6.1 commutes with the required symmetries and preserves clustering and positivity, with the same form of growth. This proves applicability for either centering.

On the cyclic domain, define Φ^0 by the displayed subtraction. Expanding products with the identity shows that the cyclic polynomial domains of Φ and Φ^0 coincide. The scalar integral is invariant under the Poincaré action on tests, and the subtraction preserves Hermiticity and locality. The vacuum and its translation representation are unchanged. The same finite subset expansion transfers its cluster property.

For strictly time-ordered distinct Euclidean arguments, the analytic correlations of the shifted field are obtained by the finite subset expansion of its Wightman products. Each retained subsequence stays in the ordered tube: the imaginary difference between retained adjacent points is a sum of the original forward differences. The constant terms have the same Euclidean and Minkowski values. Restriction to the Euclidean arguments therefore gives the shift eq. 6.i. Every configuration of distinct Euclidean arguments can be rotated to have distinct time coordinates, by choosing an axis away from the finitely many hyperplanes perpendicular to their pair differences. Both Euclidean restrictions are symmetric and Euclidean covariant. The equality in the ordered regions therefore identifies the shift on compact tests away from the diagonals. The continuous flat extensions identify it on $\mathcal{S}_{\neq,n}$. Uniqueness in the corrected reconstruction theorem now identifies this shifted Wightman field with the reconstruction from G^0. □

10. The scalar continuum conclusion

Theorem C (A realized nontrivial scalar phase)**.** Let $K_j \in \mathbb{Z}_{\geq 0}$ and $a_j = L^{-K_j} \downarrow 0$ give one sequence of admitted real data with bare couplings $\beta_j \geq 0$, as in section 2. Assume that the actual ordinary source, rooted-domain and insertion hypotheses of [3, Theorems B and C] hold uniformly on these data. Assume also the state-uniform physical two-point hypotheses of [4, Theorem A], using its finite-reference route or its cofinal alternative, with one common source normalization and with the comparison applicable eventually at each fixed sufficiently small physical separation. Use the external plateau and spectral-carrier estimates stated in section 7 and proposition 8.1 and the four analytic criteria stated in section 8.

Choose reference states $\nu_j \in \mathcal{G}_{a_j,\beta_j}$, for example fixed-spacing infinite-volume torus limits. There is one subsequence with the following properties. Its self-centered reference moments converge on every separated compact C^1 tuple to a hierarchy G. There is a compact metrizable full-support phase law $(\mathcal{P}, \lambda)$ as in theorem A. Every $\xi \in \mathcal{P}$ is realized along this same subsequence by translation-ergodic Wilson Gibbs states $\rho_{j,\xi}$, with the original reference subtraction ν_j.

The full phase moments $G_{\xi,n}$ extend uniquely to $\mathcal{S}_{\neq,n}$. Their degree-zero value is one and, with constants $1 \leq A, B < \infty$ independent of n and ξ,

$$|G_{\xi,n}(\varphi)| \leq AB^n (n!)^{k+32} |\varphi|_{20n}, \qquad \varphi \in \mathcal{S}_{\neq,n}. \tag{10.i}$$

Here k is the fixed exponent in eq. 2.iv; the degree-zero convention is that of definition 2.2. The same estimate holds for the self-centered phase hierarchies and the reference mixture after enlarging A, B. For every fixed flat test φ, the reference-centered evaluations $\xi \mapsto G_{\xi,n}(\varphi)$ are continuous, and their λ-integrals equal the corresponding moments of the reference mixture.

Each phase is real, permutation symmetric and Euclidean invariant on these test spaces. It is reflection positive on every corresponding affine positive-time flat domain. It clusters on separated compact groups, and its ordered Schwartz sequences satisfy the spatial cluster condition. Its first moment is $b(\xi) \int f$, where b is a bounded continuous real function with $\int b \, d\lambda = 0$. The reference subtraction is retained in this formula.

The full reflection Hilbert spaces have the zero-energy and spatial rotation properties of proposition 8.3.

The connected two-point functions satisfy the closed law eq. 6.ii and the bounds eq. 6.iii on the full shape class of corollary 6.2, uniformly in ξ. For the fixed unit-mass positive-time profile f_d in corollary 6.3, the connected covariance in every phase satisfies

$$c \, \frac{d^{-8}}{[\log(1/d)]^2} \leq G_{\xi,2}(\theta f_d, f_d) - G_{\xi,1}(\theta f_d) G_{\xi,1}(f_d) \leq C \, \frac{d^{-8}}{[\log(1/d)]^2} \tag{10.ii}$$

for sufficiently small $d > 0$, with positive constants and a threshold independent of ξ. The corrected scalar reconstruction gives each phase a local Hermitian scalar Wightman field with positive energy and a unique vacuum in its cyclic realization. This field is nontrivial. ■□

Proof. At each fixed datum proposition 2.1 gives the required reference states. The actual source hypotheses give eq. 2.iv for every integrating/reference pair, by [3, Theorem B]. Apply proposition 3.1 to the pair (ν_j, ν_j). This chooses the single subsequence used from now on. The reference singleton is zero. The phase construction in theorem A uses the uniform bound before passing to the limit. It supplies the realizing states on that same subsequence and the uniform compact cluster law.

Apply proposition 4.1 with separation exponent five to each realized hierarchy. The constants depend on the common radius-one bound, so they are uniform in the phase. Self-centering uses the pairs $(\rho_{j,\xi}, \rho_{j,\xi})$; it therefore has the same graded estimate before the limit. This proves eq. 10.i. Approximate a fixed flat test by finite sums of compact separated tensors in the seminorm $|\cdot|_{20n}$. The uniform bound makes the phase evaluations converge uniformly in ξ. They are therefore

continuous, and the compact barycentre identity eq. 7.ii passes to the flat test. Proposition 7.2 identifies the mean and its continuity and average.

The full reflected form follows from proposition 5.1. The insertion estimate, spectral carrier and the physical shape law meet the four analytic criteria, as checked in theorem B. Hence each phase is Euclidean invariant. Conjugating a coordinate reflection by an isometry transfers its positive form to any affine mirror; the pullback preserves the corresponding flat test domains. The ordered Schwartz cluster condition follows from the compact cluster law by the unitary translation and Hilbert approximation argument in proposition 9.1. The same compact cluster law and Euclidean invariance give the Hilbert-space conclusions by proposition 8.3.

For each fixed d, the physical two-point bounds apply to all late realizing states $\rho_{j,\xi}$. The fixed-pair limit argument of corollary 6.2 therefore gives the closed law on the full prescribed class. Its specialization in corollary 6.3 proves eq. 10.ii, with the common constants inherited from the state-uniform bounds. Finally proposition 9.1 matches the quantitative growth, ordered positivity, symmetry and cluster assumptions of the corrected theorem. The positive connected norm excludes a scalar field proportional to the identity. □

The theorem is a deduction from the stated uniform lattice estimates. It neither chooses contact terms in the Euclidean moments nor identifies a complete algebra of Yang–Mills observables. Such an identification requires compatible joint continuum limits for additional gauge-invariant observables. Nontrivial scalar correlations alone also do not establish an interaction criterion for an underlying field theory. If Z is a standard Gaussian random variable, then $Z^2 - 1$ has cumulants $2^{n-1}(n-1)!$ at every order $n \geq 2$, as follows by differentiating $-t - \frac{1}{2}\log(1-2t)$ at zero. Thus a quadratic observable of a Gaussian variable already has nonzero higher connected moments.

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